MISR LANGUAGE SCHOOLS
National Primary Stage

Grade 6

Wonders

Leveled Readers



Name: _____

Class: _____

2026-2027

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Realistic Fiction

Silver Linings

by Susan Paris
illustrated by Soud



Mc Graw Hill Education

PAIRED READ The Big Trek

STRATEGIES & SKILLS

Comprehension

Strategy: Visualize

Skill: Compare and Contrast

Vocabulary

consolation, glimmer, heinous, indispensable, perception, phobic, sarcastic, threshold

Word Count: 2,438

Genre Realistic Fiction

Essential Question

How do new experiences offer new perspectives?

Silver Linings

by Susan Paris
illustrated by Soud

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Chapter 1 Clouds

"Every cloud has a silver lining." It drives me crazy when Mom says nonsense like that. I'd barely had time to hobble to the car and arrange myself in the passenger seat, my throbbing foot already blown up like a balloon when she comes out with it, exactly as I knew she would. First, she pats my leg, and then she just can't help herself: "Every cloud has a silver lining, you know, honey."

"Mom," I grumble, "I appreciate the thought, but I'm not ready to look on the bright side of this debacle."

She apologizes, and I can tell she means it, but it's pretty obvious her timing is way off; we haven't even left the school parking lot yet.

On the farthest side of the field, I can see my teammates doing sprints as they train. As I continue watching, Coach Zito strides back toward them. He'd mercifully helped me to the car in silence. Unlike Mom, he knew when it was best to say nothing—he knew what was at stake.



As Mom starts the engine, I imagine my teammates cooling down without me while they listen to one of Coach Zito's pep talks. And next Thursday, they'll go to practice without me; and the Thursday after that …

When I had fallen and twisted my ankle, I'd cried from the pain, but now I'm crying out of self-pity. Hours and hours of training, and now this. I tell myself there's a glimmer of hope that everything will turn out okay, that I'll make it to the track-and-field meet next month, but I know it's just that: a glimmer.

The tears fall hot and fast; I drag the back of my hand under my nose and sniff loudly. Mom gives me a weak smile. "It'll be okay, Casey. You'll see."

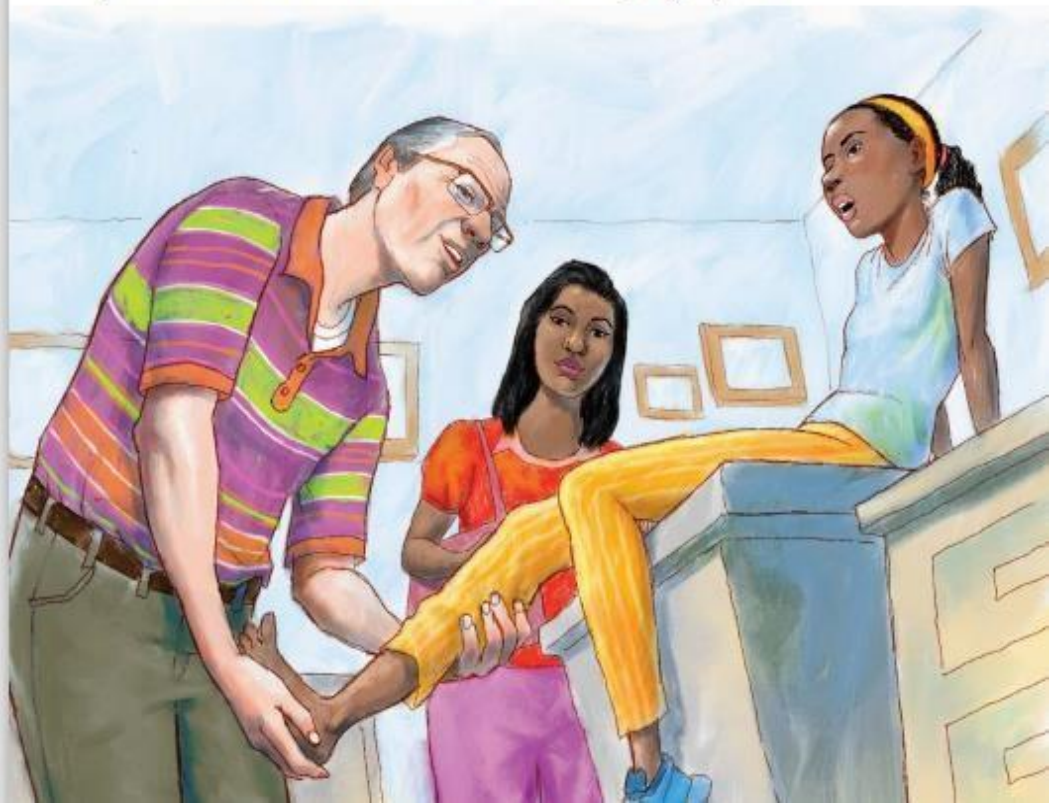
I suspect this is another version of the silver-lining thing, but I feel so miserable that I let it pass. I mean, what's the point?

Mom has already called the medical center, so we don't have to wait to be seen. Doctor Van Dyk asks me some questions and then gently picks up my foot and moves my ankle around a bit before carefully lowering it.

I get straight down to business and nervously ask how long it will be before I can run.

"Four weeks maybe, possibly five," he answers, "and even then, you'll need to take it easy for a while. Is that possible?" he asks with a smile. He knows me well.

I can't answer. My eyes fill with tears, my throat begins to tighten, and I have to focus on his brightly patterned shirt so that I don't start crying again.



I'm exhausted, and it's a relief to get home. Chris must have heard the keys in the front door because he comes flying down the stairs two at a time, a habit that always irritates Mom. She's warned him hundreds of times not to do that because he could … sprain an ankle!

Chris takes one look at my face, and then after glancing at Mom, he follows her pointed gaze down to my foot.

"Your ankle looks like it swallowed a tennis ball!"

"Yeah, something like that," I mutter.

"Does it hurt?"

"No. Actually, it tickles," I say. In truth, I crossed my pain threshold hours ago, but I'm not about to admit it.

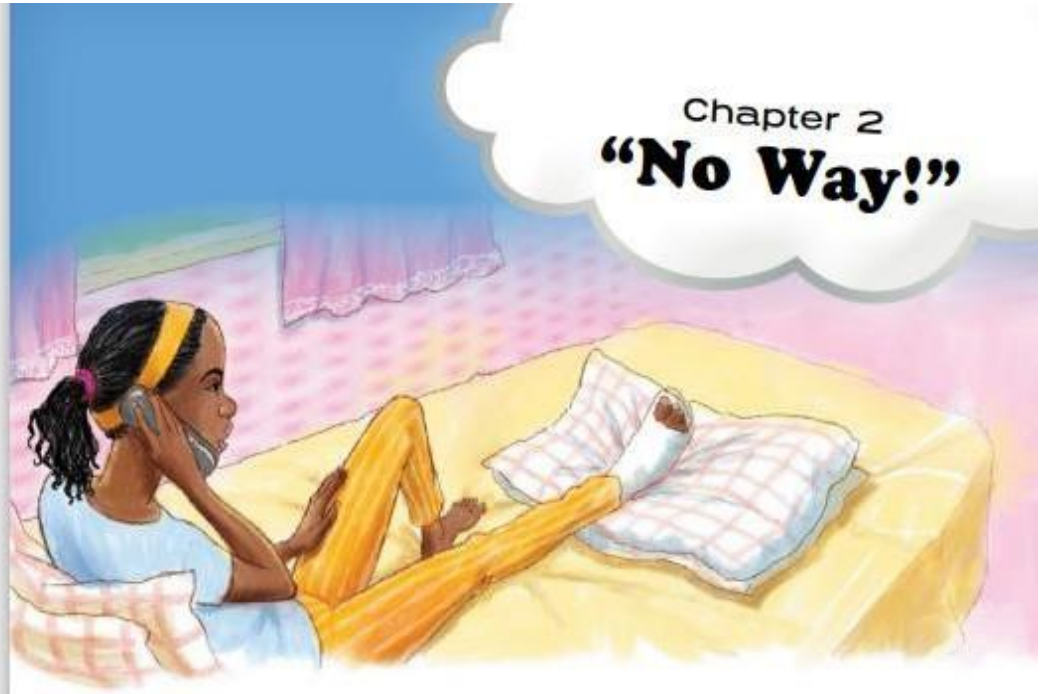
"No need to be sarcastic," Chris says, pretending his feelings have been hurt. "I was only asking." He gives me a sympathetic smile and heads off to raid the refrigerator.

I swing through to the den, thumping along like some kind of oversized, uncoordinated bug. Doctor Van Dyk said I would probably only need crutches for a week or so, but already my hands hurt and my shoulders ache.

I collapse onto the sofa, toss the crutches aside, and reach for the television remote.

STOP AND CHECK

How do the different characters in this chapter react to Casey's situation?



Chapter 2
"No Way!"

Kim calls me as soon as she hears. "You poor thing," she commiserates, "you must feel *totally* ripped off!"

Despite everything, I smile. Kim is allergic to exercise—she couldn't run a mile to save herself—but somehow she understands how much track means to me.

We spend the next ten minutes discussing my rotten luck, then Kim says something completely unexpected.

"Maybe this is a chance to try something new," she suggests. "You know, silver linings and all that."

My eyes instantly narrow with suspicion: "Have you been talking to my mom?"

"No … What are you talking about?"

From her tone, I can tell she's innocent. "It doesn't matter. What were you going to say?"

"Well, we need help with the school play."

"The play? No way! That's so not my scene. Plus, I'm on crutches, remember? Not to mention, I'm a terrible actor!"

"I wasn't thinking of acting. We need a prompter, and they just sit around—perfect!" Kim replies.

"Sitting around doesn't really sound like me, either."

"True," Kim says with a laugh, "but please please give it a try. We're desperate, and I could really use my best friend's help."

It's emotional blackmail, and Kim knows it. I don't know why I'm so reluctant; maybe it's because I'm not ready to accept that I won't be competing at the track meet. But it's more than that: I'm just not what teachers call a "team player." While other kids seem to love doing stuff together, that's just not me.

"Casey?" Kim's voice reminds me that while I'm not much of a people person, I still need her in my life.

I heave a dramatic sigh and ask, "So, what does a prompter do, anyway?"

Less than 24 hours later, Kim is guiding me down the stairway of the darkened auditorium.

I look around at all the kids talking and laughing and tell Kim I'm feeling kind of phobic.

"Phobic?" Kim asks. "What is there to be scared of?"

I point at all the people and explain that I'm used to being in the open air. Why is there such a crowd?

"It's a play, silly, it doesn't happen on its own." And she drags me over to introduce me to Mr. Leshko, the director.

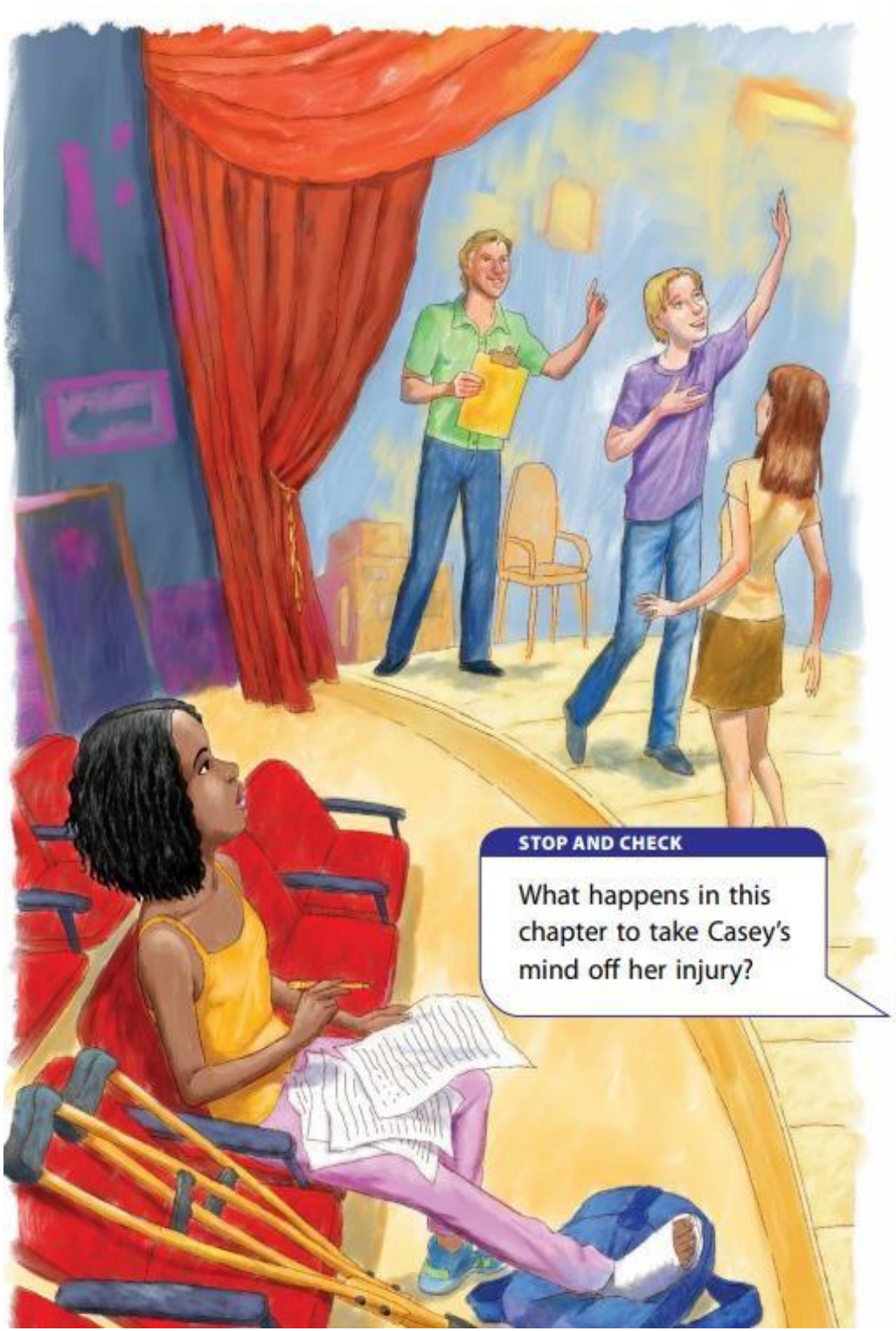
Mr. Leshko seems grateful for my help. He gives me a script, explains what to do, and then leaves me to it. So does Kim, who disappears off to the dressing rooms.

It turns out that being the prompter isn't as heinous as I'd thought it might be. Basically, all I have to do is listen carefully and remind people of their lines when they forget them. I pick up the role pretty quickly. And I soon see why they need me. The actors mess up all the time, and I'm yelling out so many lines, I have to ask for a glass of water. The actors must be used to it though because they don't seem to mind.

Kim comes by later to ask how it went.

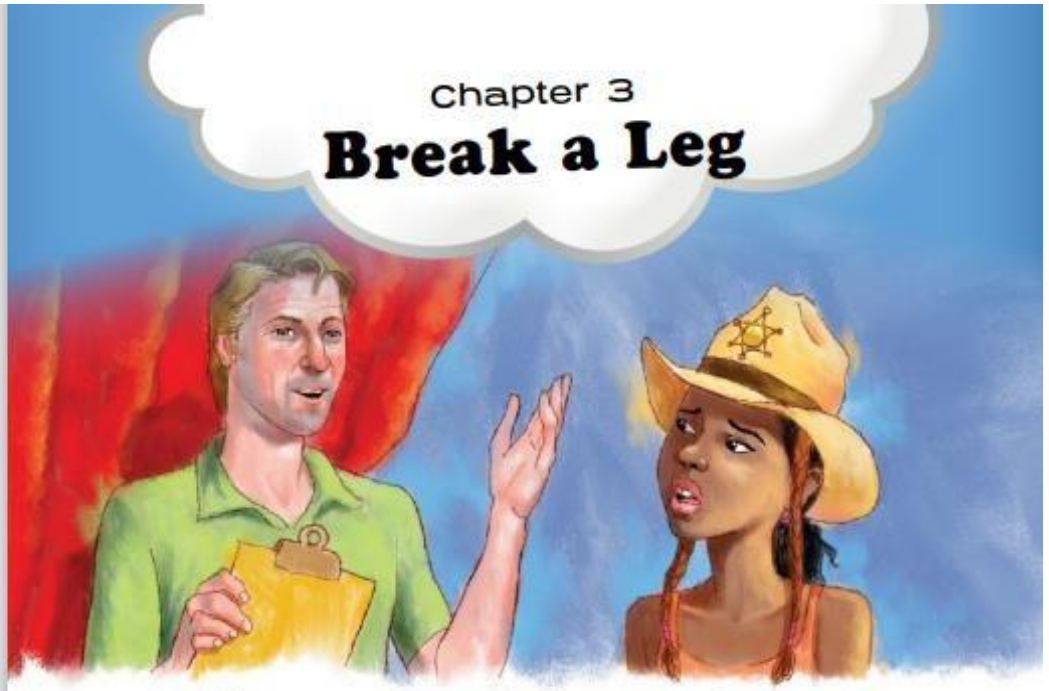
"Pretty well," I admit, "but I think they need more training. They forgot tons of lines!"

"Actors don't train, they rehearse," she corrects.



Chapter 3

Break a Leg



The following week, everybody seems to be coming down with the flu, which is how I get conned again.

"Casey," Mr. Leshko says, "Amanda's sick, so could you fill in for her at rehearsal today?"

"But she plays the sheriff," I squeak, "and I can't act."

He asks if I've ever tried, and I have to admit I haven't—but acting isn't my thing. Mr. Leshko just gives me an exasperated look.

I have one last chance of escape, and it's a good one: "My ankle," I say, dangling it limply between us.

"No problem. I'll get you a chair, and you can pretend you're on horseback." Then he's gone, just like that.

So I play the sheriff, and because I've been the prompter, I know the lines. I even do a pretty good job.

Mr. Leshko notices. After rehearsal, he walks over. "Good work, Casey. Did you enjoy yourself?"

And I have to confess, it was okay.

Dad picks me up and drives me home past the running track. Feelings of frustration and disappointment overwhelm me. The team will begin training for the 1500 meters soon—my best event. Last year, I came in second, while this year, I'll come in nowhere.

Dad senses my gloomy thoughts, and lowering the volume on the radio, he asks when the play opens.

"In three weeks, but lots of the actors are still messing up their lines, so it could be a disaster."

"I don't know; a lot can happen in three weeks. And why are you being so negative, anyway?"

"Because I'm not into acting. What's the point?"

Dad is on to me immediately. "You mean what's the point when there are no winners or losers?"

"Well, at least when I train, I get results. The kids who are trying to act are still making mistakes."

"Maybe they're simply having fun," Dad suggests. "Life is not all about winning prizes, you know."

"Huh!" I mutter—as if having fun is any consolation for missing out on competing in the track-and-field meet.

Amanda misses the next rehearsal and then the one after that too. I progress from crutches to an ankle brace and can move around the stage, and Mr. Leshko starts giving me directions. Kim thinks it's hysterical.

"You know," Kim comments after my fifth time filling in for Amanda, "you're actually really good." And she warns me to watch out. "Amanda's not indispensable," she cautions. "You're just as talented as she is."

Has Mr. Leshko been waiting in the wings for just this moment? Suddenly he's there, like he's personally scripted the scene himself.

"Kim's right," he says. "I need you to take the role permanently. Amanda won't be back in time."

"I can't," I remind him. "My limp …"

"I'll write it into the play," Mr. Leshko replies and asks Kim to help me practice my lines.



The sheriff is assisted by a deputy, who is played by Ethan Sosnowski. Everybody adores Ethan. But when he suggests we run through our scenes together, I'm surprised and automatically refuse.

"What is it with you?" Ethan asks, exasperated. "Oh, I forgot, you're a big track-and-field hotshot. I guess it's no fun if you're not the star of the show, huh?"

Ethan sees my face and hastily apologizes: "Sorry, Casey; I didn't mean it."

"That's okay," I mumble. "Most people would have the same perception of me."

He relaxes into a grin. "Probably. So do you want to practice or not, hotshot?"

Only someone totally clueless would turn him down twice, and he is kind of cute …

"Be careful, you almost looked as if you were having fun out there today," Kim teases me on the way home.

"We were only practicing our blocking," I insist weakly.

"Whatever you say!" Kim dismisses my excuse with a smile. "Only three days to go. Can you believe it?"

What I can't believe is the tingle of excitement I feel when I think about opening night.

Before our first performance, Ethan tells me to break a leg. I glare at him. Is he trying to be funny?

"It's what actors say—it's figurative," he explains. "It means good luck."

"Really?" And I start snorting with laughter. "Imagine saying 'Break a leg' to someone before the 1500 meters."

Ethan starts laughing, too, until we're both laughing so hysterically that the stage manager shoots us a dirty look.

I hate waiting, and I confess that I'm feeling nervous.

"You don't look nervous to me," Ethan observes, and then he gently pushes me onto the stage and into the glaring brightness of the spotlights.



I'm not sure who is the most enthusiastic about opening night: Mr. Leshko or my family.

"You all looked so *relaxed*." Mr. Leshko nods approvingly, and I can tell he's impressed.

"When did you learn to lasso like that?" Dad asks.

"Stop it, everybody," interrupts Chris, "or she'll get a swollen head to match her swollen ankle."

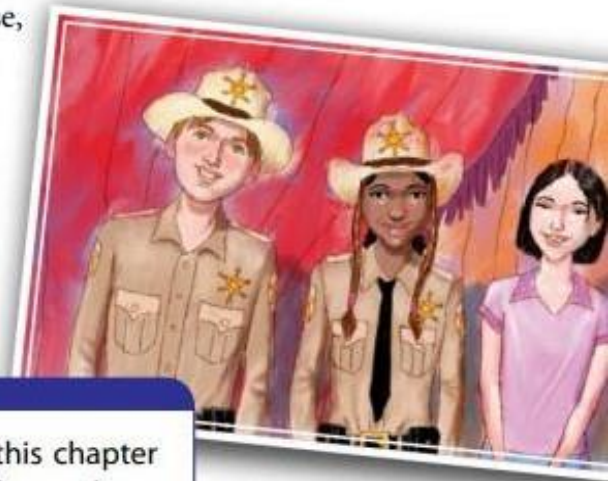
"Chris!" says Mom. "Well, I think it's great that Casey is learning she has more talents than just running."

"And the best part is that we get to do it again tomorrow night," Ethan adds. "And every night this week!"

"True." I smile. I hadn't thought of that.

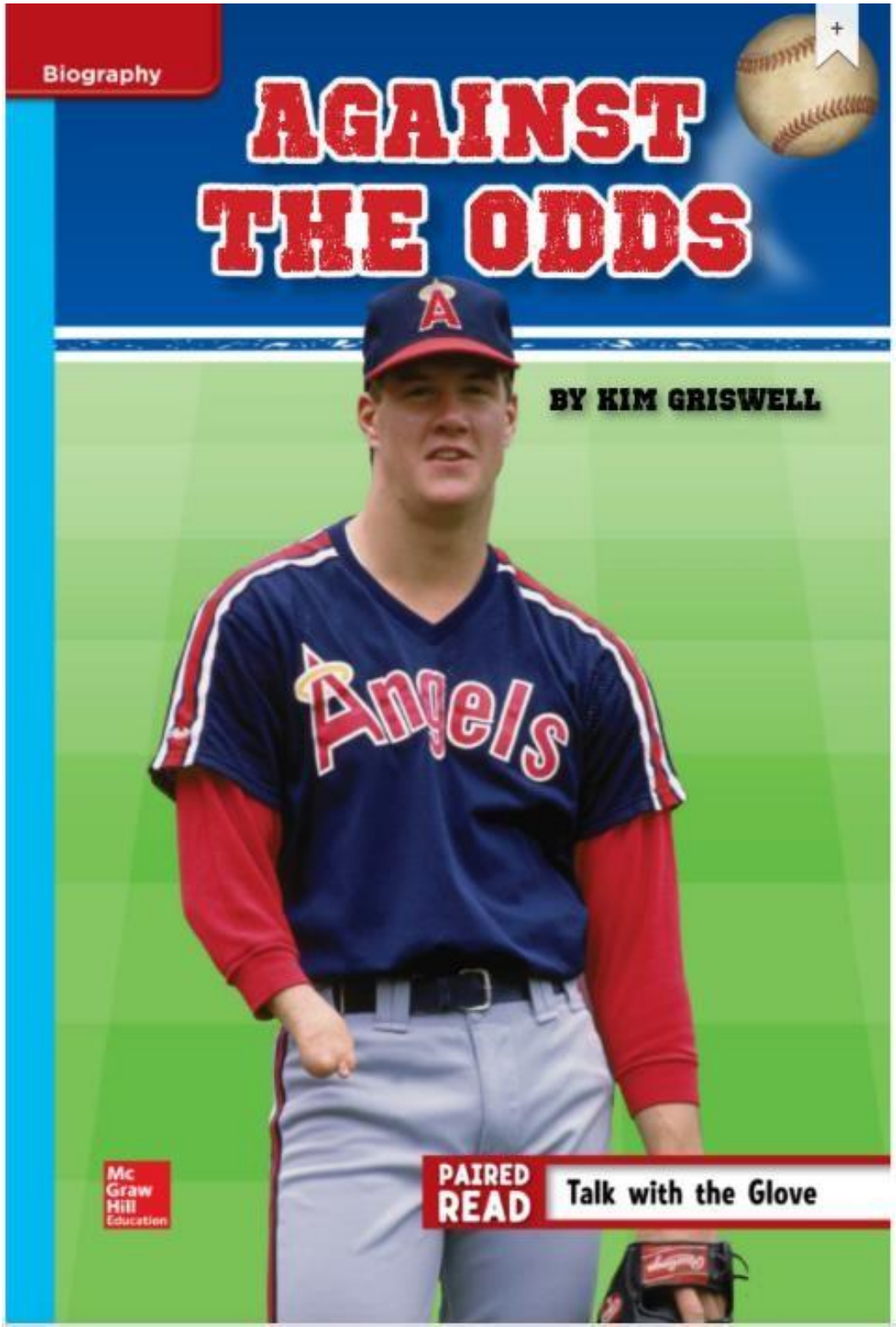
"Hey, look, she's smiling!" Chris says. "Does anyone have a camera to record this momentous event?"

Dad does, of course, and we line up: Kim on one side of me, Ethan on the other. Maybe being part of a group isn't as scary as I'd thought.



STOP AND CHECK

How do the events in this chapter show how Casey has changed since the beginning of the story?



Genre Biography

Essential Question

How do people meet personal challenges?



AGAINST THE ODDS

BY KIM GRISWELL

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Image Source/Getty Images

CHAPTER 1

A WAY TO PLAY

Jim Abbott grew up in Flint, Michigan. He was born in 1967 and by the age of five, he already knew he wanted to play baseball. But when the neighborhood kids were starting a game, they didn't ask him to play.

Jim ran home in tears. His parents were sympathetic but firm. Mike and Kathy Abbott knew that if their son wanted to live up to his potential, he had to be brave. Jim had been born without a right hand.

Jim's parents told him to go back and tell the other kids he wanted to play. He followed their advice and got into the game. But he knew that if he wanted to stay in the game, he'd have to do with one hand what others did with two.

With the help of his dad, Jim found his own way to handle a baseball. To compensate for not having a right hand, he learned to throw and catch with his left. He called his technique the "glove-hand switch."



Playing in Little League games is something that many young children look forward to doing.

Jim also had to learn to swing a bat with one hand. He did this by holding the bat with his left hand and balancing the handle on the stub of his right arm. Most of the time, he connected with the ball. Jim's parents had taught him to work with what he had, rather than focusing on what he did not have. This was an important trait that helped him to overcome his personal challenges.

Jim joined Little League at age 11. By the fifth inning of his first game, his team was so far ahead that the coaches invoked the mercy rule and ended the game. He had just pitched his first no-hitter.

THE GLOVE-HAND SWITCH

Fans loved to watch Jim Abbott's famous glove-hand switch. He moved so fast, it was hard to see what he was actually doing. These are the moves:

1. **Abbott throws the baseball with his left hand, keeping a fielder's glove balanced on the end of his right arm.**
2. **As soon as he releases the ball, he switches the glove onto his left hand.**
3. **After he catches the ball, Abbott cradles the glove in the crook of his right arm as he pulls his left hand out of the glove.**
4. **He takes the ball out of the glove and gets ready to throw again.**

By the time Jim entered junior high school, he had an 80-mile-per-hour fastball. Other pitchers his age threw at speeds of 50–60 miles per hour. In high school, he quickly earned a spot on the varsity team.

Some opposing coaches thought fielding a bunt would be Jim's weakness. Bunts are short-hit, infield balls that the pitcher often must field by running from the **pitcher's mound**. One coach decided to implement a strategy to test this idea. He told eight batters in a row to bunt Jim's pitches. He thought Jim's glove-hand switch would slow him down in fielding the bunts. The first batter to bunt made it to base, but Jim retired the next seven, one after another.



When a bunt is played, the catcher usually runs in to field the ball.

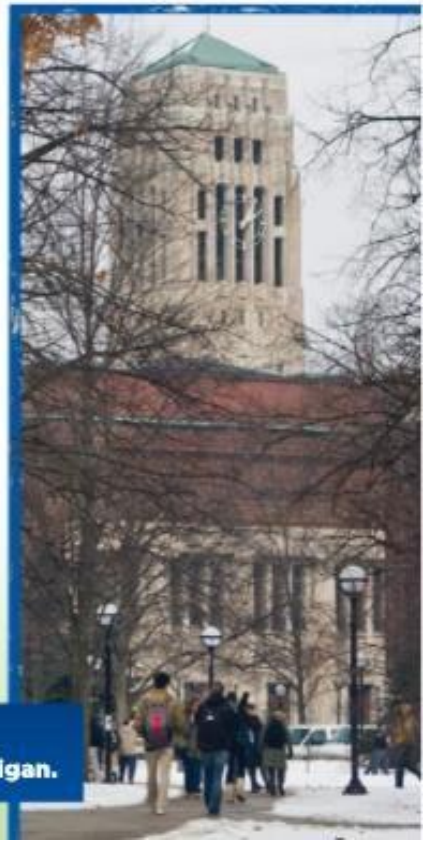
Jim's left arm kept getting stronger, and that made him a great batter and an incredible pitcher. Batting from the left side, he produced enough power to blast seven home runs out of the ballpark in his senior year. His team won ten games and lost only three. Jim averaged more than two strikeouts per inning.

CHAPTER 2
AGAINST THE ODDS

Jim Abbott's fantastic senior baseball season caught the eye of a professional team, the Toronto Blue Jays. He had a big decision to make: go to college or head straight for the pros. His parents encouraged him to think beyond baseball and pursue an education.

The University of Michigan offered Abbott a full scholarship to play college ball in Ann Arbor. Abbott knew he had to carefully assess his options. If he took the scholarship he wouldn't be going pro, but he would be close enough to home that his parents could come to his games. Pitching for the Michigan Wolverines would also let Abbott develop as an athlete without the pressure of playing on a professional team.

Despite the $50,000 bonus the Blue Jays offered to sweeten their bargain, Abbott opted for college.



Students walk on the University of Michigan campus in Ann Arbor, Michigan.

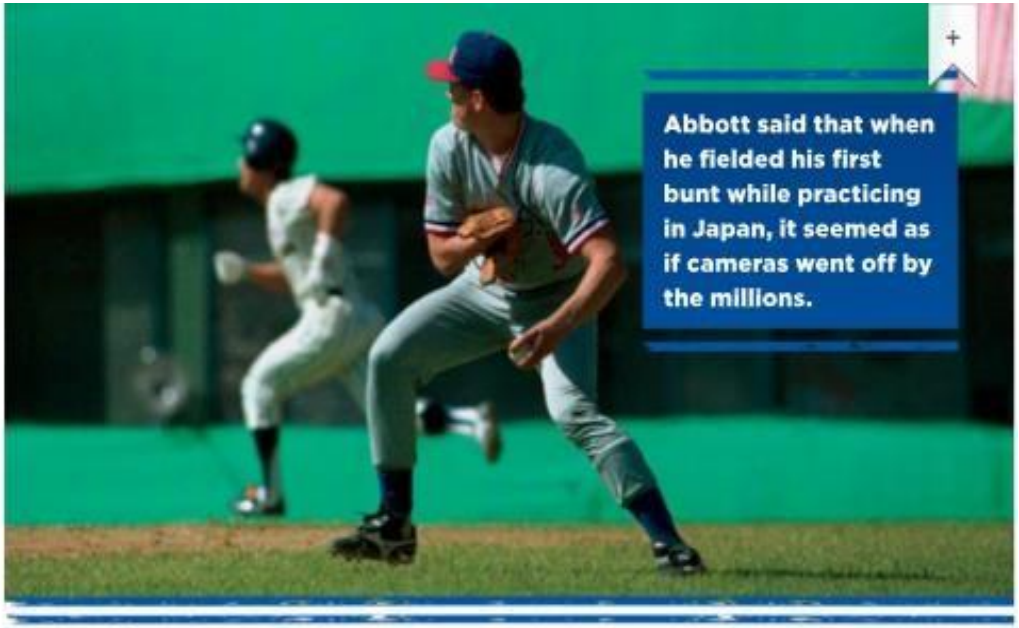
Following Abbott's first college win came a wave of national media attention, even though much of the buzz was because Abbott did not have a right hand. As far as Abbott was concerned, however, having no right hand was peripheral to his achievements. It did not define him as a pitcher. Proving this belief, Abbott closed his freshman year with a record that helped his team win the 1986 Big Ten Conference title, a prestigious college competition.

The following year was even better. Abbott earned a spot on the United States national amateur baseball team and was selected to carry the flag for Team USA into the Pan American games. During the competition, he pitched for two wins without giving up a single **earned run**. Team USA captured a silver medal.

Abbott's sophomore year culminated with his being chosen as the first baseball player to receive the Sullivan Award.

THE SULLIVAN AWARD

The Amateur Athletic Union's Sullivan Award is the top amateur sports award. It honors the most outstanding amateur athlete in the United States in a given year. For the 1987 Sullivan Award, Jim Abbott beat out basketball's David Robinson, volleyball star Karch Kiraly, and Olympic gold-medal swimmer Janet Evans. The winner is chosen not only for athletic ability but also for exhibiting the qualities of leadership, character, and sportsmanship.



Abbott said that when he fielded his first bunt while practicing in Japan, it seemed as if cameras went off by the millions.

Abbott's success brought a barrage of questions from the press. His least favorite was, "How do you play baseball with just one hand?"

"I don't even think about it," Abbott told them, "until you guys come around to remind me."

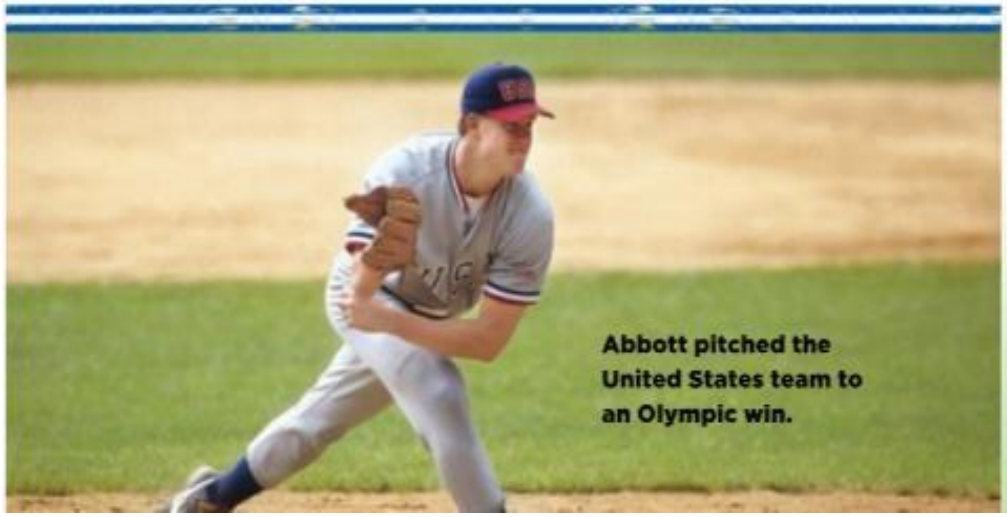
Abbott's winning streak gave him the confidence to successfully try out for the United States Olympic baseball team. In June 1988, he joined Team USA on a worldwide tour that would end at the summer Olympic Games in Seoul, South Korea.

The highlight of Abbott's summer was the **series** against Japan's national team. The Japanese were fascinated with his one-handed fielding technique, and when Abbott fielded a bunt at his team's first practice, the press went wild. By the end of the week, he'd become a national hero in Japan.

By the time they reached Seoul, Abbott had become Team USA's top starting pitcher. The first game began on September 19, 1988, Abbott's twenty-first birthday. Team USA's coach, Mark Marquess, kept Abbott off the field for all but one first-round game. He had decided to save his best left-hander for the medal round.

The final game pitted the United States against Japan. Abbott started out strong, but in the bottom of the sixth, his pitching deteriorated. Two batters had made it to base with no outs. Coach Marquess had to make a choice: keep Abbott on the mound, or send him back to the **dugout**.

Coach Marquess had a feeling that his top pitcher could still turn things around. Abbott did. He led the U.S. to a Gold Medal, 5-3 victory over Japan. It was the first gold medal win in Olympic Baseball competition for the United States.



Abbott pitched the United States team to an Olympic win.

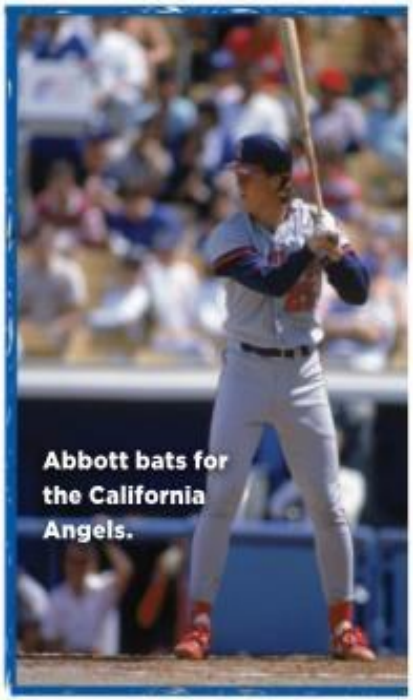
David E. Klutho/Sports Illustrated/Getty Images

CHAPTER 3
INTO THE MAJORS

Abbott returned to the United States primed to play for the major leagues. The California Angels had selected him as their first-round **draft** choice.

Everyone thought Abbott would spend time in the minor leagues, which would allow him to become a more seasoned pitcher before moving up to the team's major league roster.

That didn't stop the media from swarming the field for his **spring training** debut in Arizona. On March 3, 1989, Abbott was due to pitch against the San Diego Padres. So many reporters and fans showed up that the game was moved from the practice field to the regular stadium. Abbott gave them a show, pitching for three innings and striking out four batters.



Abbott bats for the California Angels.

When reporters asked how he could pitch so well with just one hand, he replied, "I've been doing this since I was five years old. It's as natural as tying my shoes."

Rich Pilling/Major League Baseball/Getty Images

Before spring training ended, the Angels' pitching coach, Marcel Lachemann, came to Abbott with unexpected news. Abbott had made the A team without playing a single game in the minor leagues. Only 15 other players had gone straight to the major leagues since the establishment of the amateur draft. It was one of the happiest days of Abbott's life.

From the first time he took the mound in Anaheim Stadium, Angels fans embraced Abbott. The crowd of more than 46,000 fans stood up and cheered. And when first-game jitters made the young pitcher look like the **rookie** that he was, they didn't desert him. He walked off the mound at the end of the game with his first loss in the major leagues, a 7–0 defeat. The fans gave him a second standing ovation.

Abbott wished he'd never received that second ovation. He didn't want to be a star because the media had turned him into a human-interest story. He wanted to be a good pitcher.



Otto Greule Jr/Stringer/Getty Images Sport/Getty Images

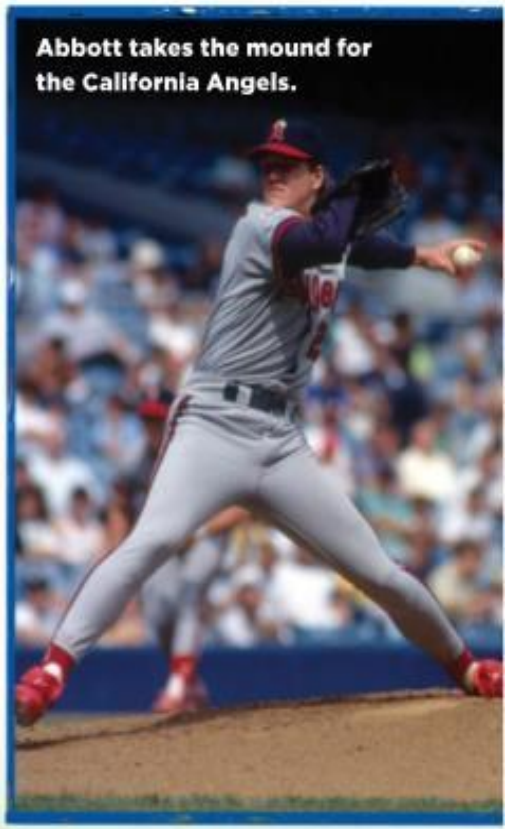
Abbott went straight to the major leagues with the California Angels.

As anyone would, Abbott experienced self-doubt after his first loss in the major league. He knew he would have to go the extra mile to succeed in the game.

That perseverance paid off. His third start as a pitcher ended with a win over the Baltimore Orioles. With renewed confidence, Abbott racked up 12 wins before the end of the season. He had set another record, the most wins for a major league pitcher in his first season since 1924.

Abbott's pitching continued to improve. By the end of the 1991 season, he had an 18–11 win–loss record and a 2.89 Earned Run Average. The Angels upped his salary to $1.85 million. This made Abbott the highest-paid fourth-year pitcher in baseball history.

In December 1992, the Angels traded Abbott to the New York Yankees. It was to be one of the most difficult—and potentially devastating—transitions of Abbott's career, but he was determined to turn disappointment into opportunity.



Abbott takes the mound for the California Angels.

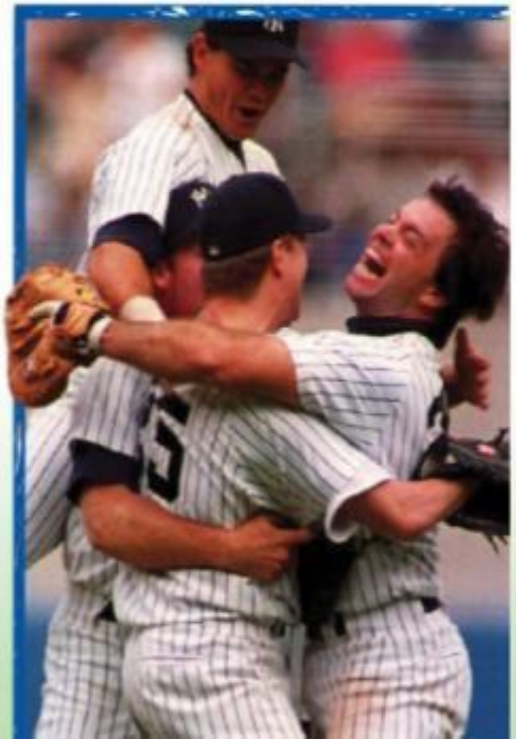
CHAPTER 4

THE PUBLIC STAGE

Abbott handled the move to New York by delivering a game for the record books. By September 1993, the Yankees were in a **pennant race** against the Cleveland Indians.

On September 4, the coach put Abbott on the mound. By the sixth inning, the Indians still had no runs on the board, and the Yankee fans were on their feet cheering. Jim Abbott was six innings into pitching a no-hitter.

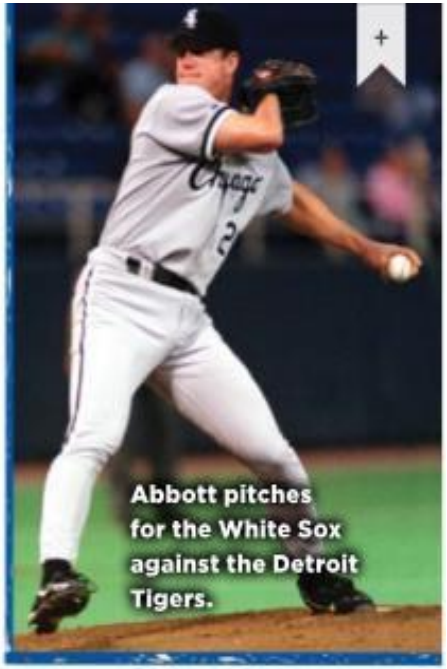
In the last two innings, the crowd stood for every pitch. Abbott felt tension mounting all around him, but he remained calm. With fans screaming in the stands, Abbott retired batter after batter. By the end of the game, he had accomplished what no other Yankee pitcher had done in a decade. He had pitched a no-hitter.



Yankee team members congratulate Abbott on pitching a no-hitter.

Abbott had reached his summit. The no-hitter shaped his legacy, putting him into an exclusive club with baseball's best pitchers, such as Sandy Koufax, Bob Gibson, Nolan Ryan, and Tom Seaver.

It also put him back on the front page, where the media spotlight's glare became harsher. On the mound, his pitching lost steam.



Abbott pitches for the White Sox against the Detroit Tigers.

Yankees' owner, George Steinbrenner, blamed Abbott's work with charities and disabled children for the slump. He accused him of not giving baseball 100 percent. Abbott met with at least one disabled child during each of his road series. His charity work earned him the Freedom Forum's Free Spirit Award. The work was important to him, but so was baseball.

When the Yankees failed to sign Abbott for the next season, he signed with the Chicago White Sox before heading back to California in 1996. It was a disaster. His fastball had gone from being clocked at 94 miles per hour down to the mid-80s. This was the speed he'd thrown back in college.

The Angels released Abbott the following spring. He overcame his disappointment and worked his way back into the major leagues for two more seasons before finally retiring in 1999.

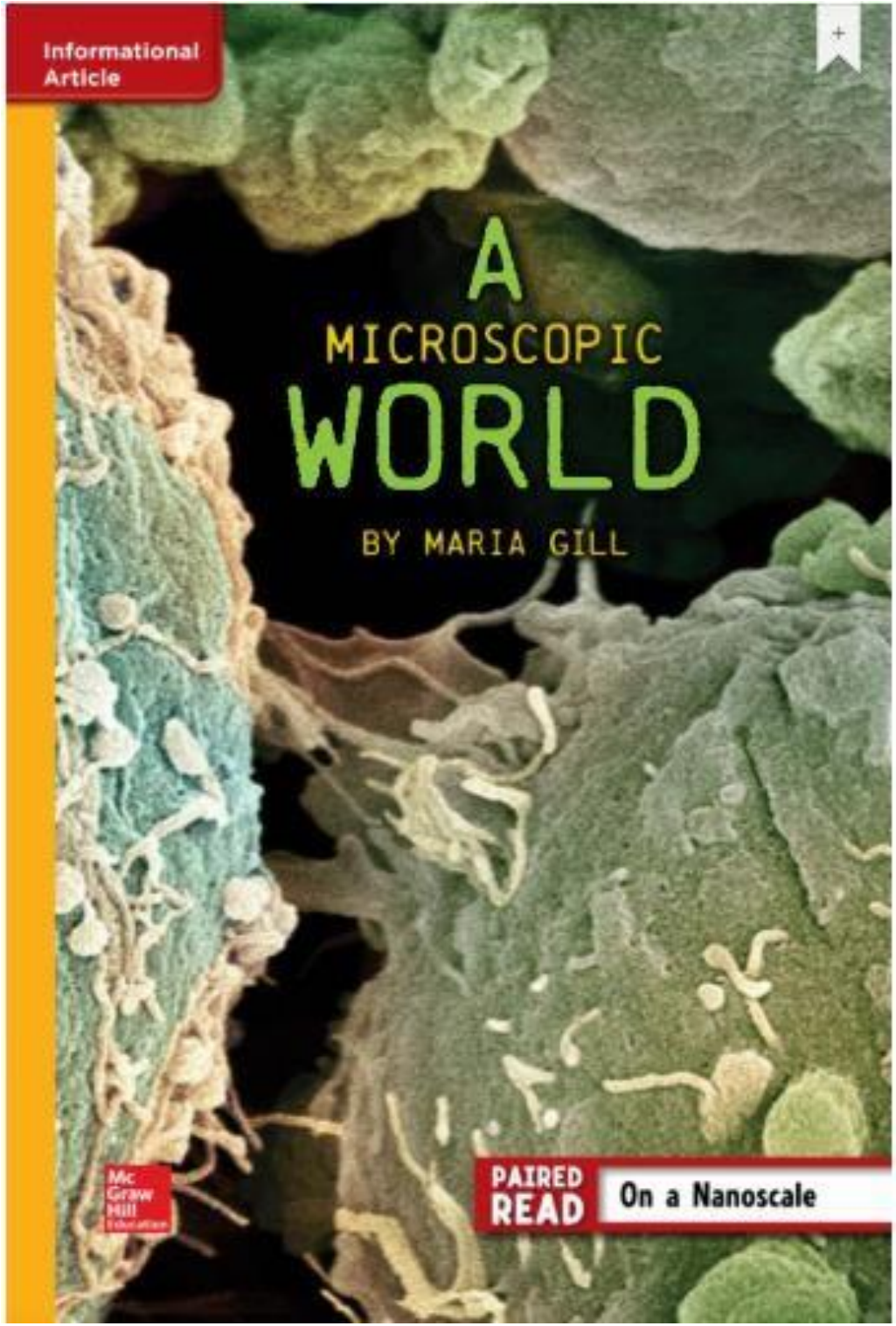
Jim Abbott left the spotlight of professional sports behind, but not the public stage. He now spends his days encouraging people to identify what they want to do, focus on their abilities, persevere, and go after their dreams, just as he did.

JIM ABBOTT'S CAREER PITCHING STATISTICS

Year	Team	W	SO
1989	California Angels	12	115
1990	California Angels	10	105
1991	California Angels	18	158
1992	California Angels	7	130
1993	New York Yankees	11	95
1994	New York Yankees	9	90
1995	Chicago White Sox	6	45
1995	California Angels	5	41
1996	California Angels	2	58
1998	Chicago White Sox	5	14
1999	Milwaukee Brewers	2	37

W–Win: number of games where pitcher was pitching while his team took the lead and went on to win.

SO–Strikeout: occurs when a batter receives three strikes during his time at bat. Pitchers want to throw as many strikeouts as possible.



VOCABULARY & SKILLS

Comprehension Skill
Author's Point of View

Vocabulary
applications, catastrophic, computations, deployed, elevated, magnetic, obsolete, subsequently

Expand Vocabulary
advances, convert, enhancement, image, model, solution, vital

Content Standards
Science
Science as Inquiry

Word Count: 1,539**

Photography Credit: Steve Gschmeissner/Science Photo Library RF/Getty Images

**This total word count is based on words in the running text and headings only. Numerals and words in captions, labels, diagrams, charts, and sidebars are not included.

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Genre Informational Article

Essential Question
How have tools used for exploration evolved over time?

A MICROSCOPIC WORLD

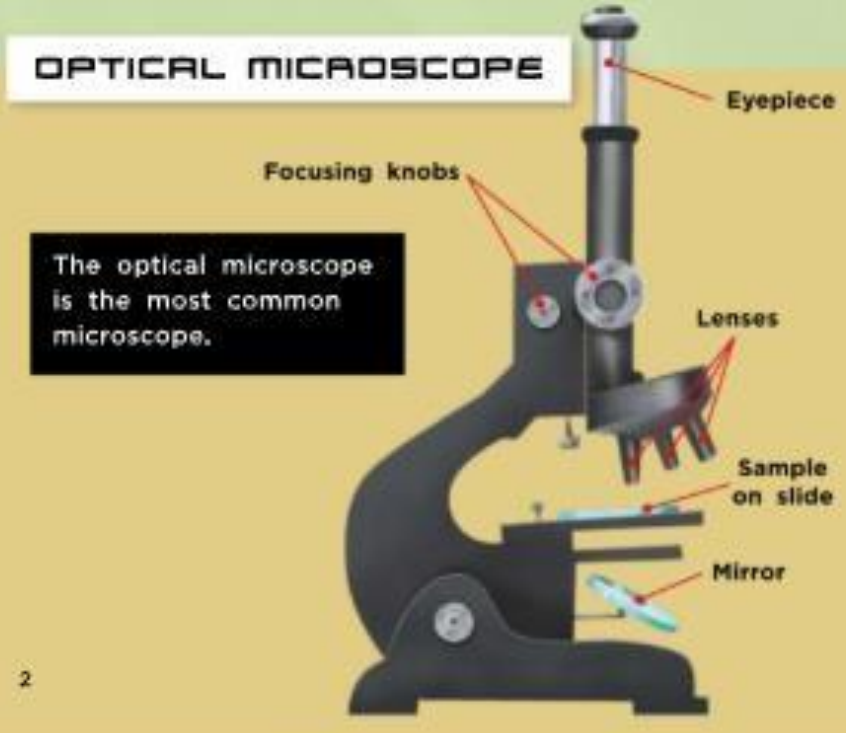
BY MARIA GILL

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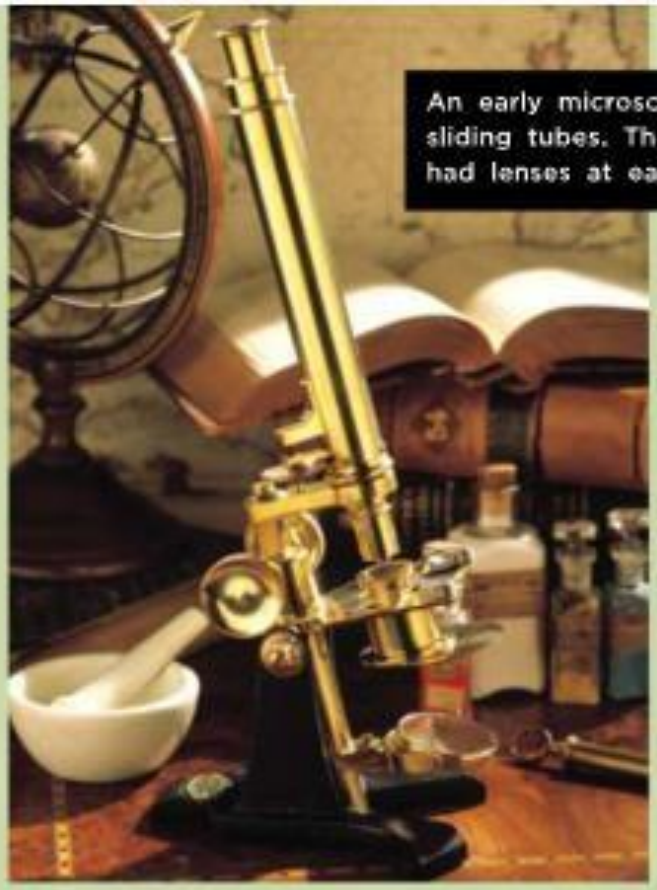
INTRODUCTION

Look at the pad of your thumb. If you used a magnifying glass, you could see the pattern of your thumbprint. With a microscope, you could see much more. An early **model**, or version, of a microscope lets you see your skin **cells**. One of the latest microscopes lets you see a single **atom** in one of those cells.

Microscopes have opened up a whole new world to us.



The optical microscope is the most common microscope.



An early microscope used sliding tubes. The tubes had lenses at each end.

Microscopes magnify objects, so the objects appear larger. This lets scientists study tiny objects that can't be seen by the naked eye.

The first microscopes were invented in the 1500s. From these early models to the powerful **nanoscopes** of today, microscopes have changed our lives. Welcome to the microscopic world!

CHAPTER 1

DISCOVERING A WHOLE NEW WORLD

Around the year 100, the Romans created the first lenses. Lenses are curved pieces of glass that either focus or **disperse** light.

A beam of light slows down when it moves from air to glass. At the same time, the beam bends. As a result, an object seen through the glass is magnified. This process is called refraction.

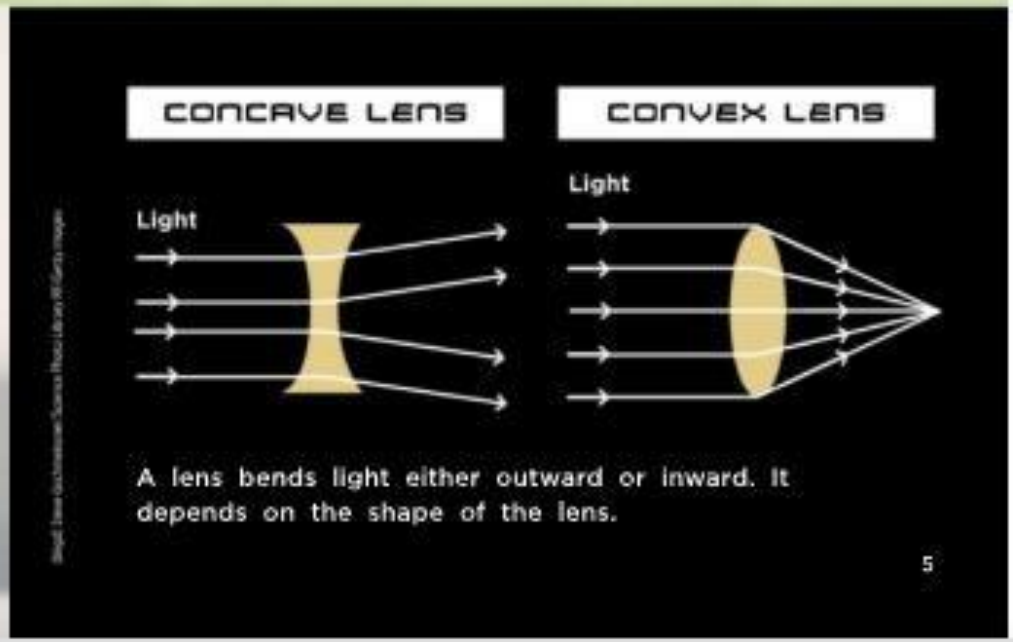


Because of refraction, a pencil in a glass of water looks bent.

In the 1590s, a Dutch lens maker and his son put several lenses in a tube. Looking through the tube made objects appear much larger. The men had invented the compound microscope. A compound microscope uses two or more lenses.

CONCAVE AND CONVEX LENSES

By the 1500s, inventors had developed lenses in different shapes. A concave lens is thicker at the edges. It helps you to see things that are far away. A convex lens is thicker in the middle. It magnifies things that are nearer.



A lens bends light either outward or inward. It depends on the shape of the lens.

DISCOVERING CELLS

Subsequently, microscopes were **deployed**, or used, for scientific research. This led to many important **advances**.

One of these steps forward in knowledge happened in the seventeenth century. A scientist named Robert Hooke looked at a piece of cork through a microscope. He saw that the cork was made up of tiny boxes, which he called cells. This was the first step toward **cell theory**.

Hooke also used a microscope to measure small objects. He placed an object and a ruler under a microscope. Then he used mathematical **computations** to figure out the size of the object.



Robert Hooke drew cork cells.

MORE MAGNIFICATION

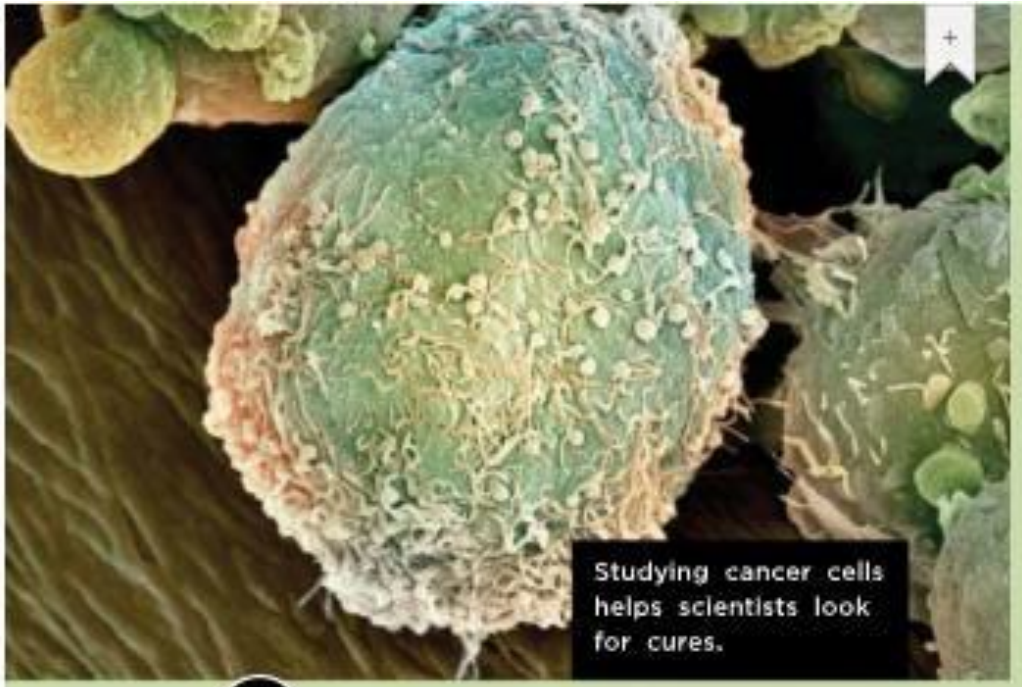
In the 1600s, a scientist named Antony van Leeuwenhoek (*LAY-vuhn-hook*) developed new microscopes. He made the lenses by grinding the glass into the shape he wanted. This made lenses that were much more powerful.

Van Leeuwenhoek's microscopes helped him make important discoveries. He identified red blood cells. He also described the life cycles of fleas and ants.

However, there were some problems with these early microscope lenses. They made objects appear fuzzy with colored edges. Many scientists didn't trust information learned from using microscopes.



Antony van Leeuwenhoek used a microscope to study bugs' eyes.



Studying cancer cells helps scientists look for cures.

BETTER LENSES

Scientists needed better microscope lenses. In the 1800s, they made improvements to lenses.

Using purer glass in the lenses made the images clearer. Another **enhancement** was the addition of the eyepiece. An eyepiece is a lens at the top of the microscope, close to the eye.

Scientists could then study smaller objects, such as **bacteria**. This helped them to learn about diseases and develop cures.

STOP AND CHECK

Why were microscopes important to scientists?

CHAPTER 2

SHARPENING THE FOCUS

Starting in the nineteenth century, microscopes were made in factories. This meant that many more microscopes were made. Small microscopes were created so people could have their own.

In the twentieth century, inventors improved microscopes even more. They lit up **specimens** to make the images clearer. They made microscopes with two eyepieces to show objects in three dimensions (3-D). This is called stereoscopy.



Stereoscopes combine two images into one. This creates a 3-D image.

ELECTRON MICROSCOPE

By the 1920s, scientists wanted to study living cells and atoms. Microscopes were not powerful enough for this. Scientists worried that microscopes were **obsolete**.

Max Knoll and Ernst Ruska proved that microscopes were still needed. These German researchers tried magnifying objects with a beam of electrons. This worked much better than a beam of light. Knoll and Ruska had invented the electron microscope.

Electron microscopes create an **image**, or picture, of the object being studied. They can show objects at 2 million times their actual size!



By the late 1930s, scientists used electron microscopes for research.

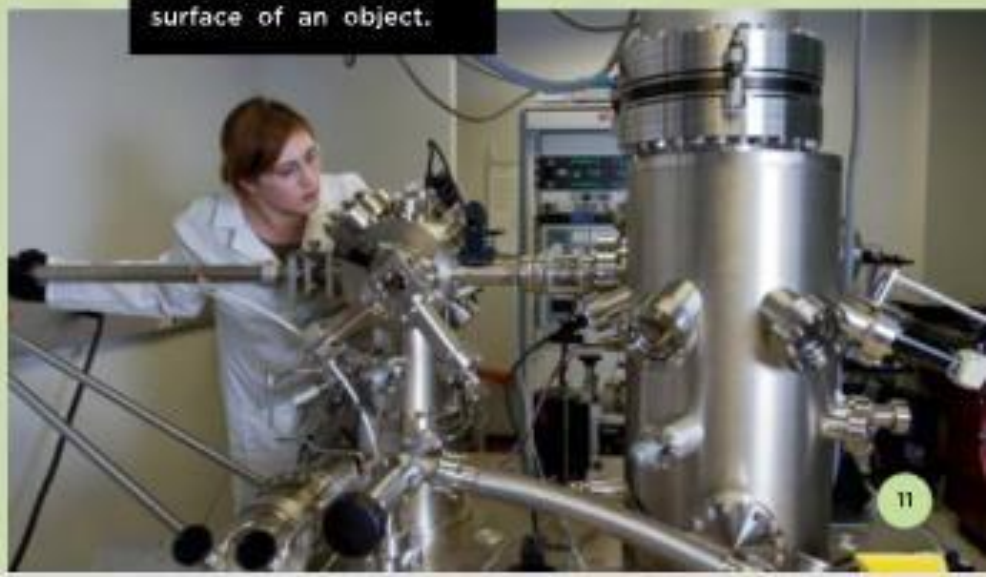
SCANNING TUNNELING MICROSCOPE

The first electron microscopes created flat, black-and-white pictures. In the early 1980s, the scanning tunneling microscope (STM) was invented. STMs create 3-D pictures. This microscope **elevated**, or increased, what microscopes could do.

STMs scan an object with a beam of electrons. This creates a map-like image. Scientists can change the colors of the image to show height, size, or shape.

The electron beams can be bent by **magnetic** fields. To stop this from happening, an STM is kept inside a strong box.

An STM can show individual atoms on the surface of an object.



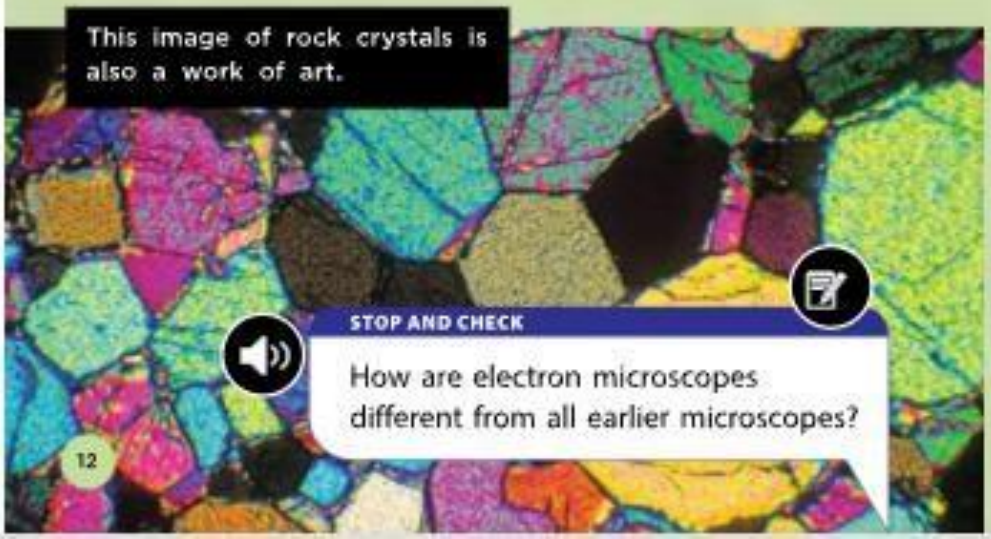
ATOMIC FORCE MICROSCOPE

STMs can scan only some kinds of materials. The materials must be able to conduct, or transmit, electricity. In 1986, the atomic force microscope (AFM) was invented. AFMs can scan more types of materials. These include glass, pottery, and biological, or living, samples.

AFMs are used in the field of nanoscience. Nanoscience is the study of atoms, **molecules**, and other tiny objects. An AFM scans an object with a laser beam.

CAPTURING MICROSCOPE IMAGES

The first scientists to use microscopes drew what they saw. When cameras were invented, scientists took pictures of what they saw. Today, scientists use computers to create digital images.



This image of rock crystals is also a work of art.

STOP AND CHECK

How are electron microscopes different from all earlier microscopes?

CHAPTER 3

NANOSCOPES

AFMs, STMs, and electron microscopes are all called nanoscopes. They can view very small objects. The objects are only nanometers (one-billionth of a meter) in size.

Scientists use nanoscopes to look for ways to decrease, or lessen, energy use. Engineers use them to design lighter, stronger materials for airplanes and cars. Nanoscopes are even used to create nanofood! One possible nanofood is tiny particles that taste like salt but have less sodium.

In medicine, nanoscopes are used to study molecules. They are also used to heal wounds. However, using nanoscience in medicine has risks. Nanoparticles can be swallowed or breathed in. They can also soak into the skin. Scientists should do more research before using nanoparticles in medicine.

SOLAR CELLS

Nanoscopes are also being used to fix environmental problems. Scientist Jennifer Dionne leads a team that researches solar cells. Solar cells **convert**, or change, light into electricity. This makes it possible to use solar power instead of other forms of energy.

Unfortunately, solar cells cannot convert all of the light. A large amount is wasted. Dionne and her team add metal nanoparticles to solar cells. This helps the solar cells convert more of the sunlight into energy.

Solar power is a wonderful alternative to other energy sources. But it is very expensive. Finding better ways to use the sun's energy is **vital**. It is the most important use of nanotechnology.



Solar cells will be less expensive if they make more energy.



Bang's cement may help to make buildings safer.

SELF-FIXING SOLUTIONS

Scientist Sookie Bang is also using nanoscopes to protect the environment. She has found a way to create cement that is good for the environment. This cement could be used to fix cracks in old buildings, such as the Empire State Building. It would be **catastrophic** if such buildings fell down. To prevent such a disaster, the buildings are constantly being repaired.

Bang uses bacteria to create the cement. The bacteria are put in sand and fed a special **solution**. When the bacteria eat the mixture, they die. This causes the sand to form into a natural type of cement.

STOP AND CHECK

How does nanoscience help the environment?

CONCLUSION

Microscopes have helped scientists make many important discoveries. Even the earliest microscopes led to big changes. They showed that doctors carried a lot of germs on their hands. Doctors started washing their hands to stop sicknesses from spreading.

Microscopes have led to other changes in medicine. One of the most important is nanosurgery. This technology has many possible **applications**, or uses. For example, doctors could use nanorobots to take medicine to exactly the right place in people's bodies. Nanorobots could also take tiny samples to test for diseases.



This is an artist's idea of a nanorobot attacking an infected cell.



This tiny microscope clips onto the back of a cell phone.

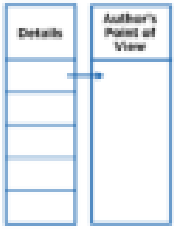
Microscopes are used for many different purposes. Some microscopes are very small. Doctors can take them along when they work away from their offices. Some microscopes are used to study atoms and molecules. Others can create 3-D images.

All of these microscopes have something in common. Each one, from the first compound microscope to the latest nanoscope, has opened the door to a new world of scientific exploration.

Respond to Reading

Summarize

Summarize how microscopes have changed over time. Use details from *A Microscopic World*. Your graphic organizer may help you.



Text Evidence

1. On page 13, where does the author express a point of view about nanotechnology?
   AUTHOR'S POINT OF VIEW

2. Find the word *decrease* on page 13. What clues help you to figure out the word's meaning?
   VOCABULARY

3. Write about the author's attitude toward microscopes. Use details from the text that show her point of view. **WRITE ABOUT READING**



STRATEGIES & SKILLS

Comprehension

Strategy: Make Predictions

Skill: Cause and Effect

Vocabulary

disposed, eavesdropping, fortitude, infinite, retaliation, rigors, stoop, undaunted

Word Count: 2,809

Genre Historical Fiction

Essential Question

How do people show inner strength?



BY TERRY MILLER SHANNON

ILLUSTRATED BY DARIA PETRILLI

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Focus on Literary Elements 20

CHAPTER 1

A SPECIAL ERRAND

Liza crawled into bed, snuggled into her pillow ... and screamed, then leaped up and clawed frantically at her cheek. Then she heard her brother and sister giggling.

"It's just a worm, scaredy-cat!" Charles sniggered.

Trying to keep her voice from trembling, Liza asked, "Why did you put it on my pillow?"

Emma laughed, picked up the worm, and carried it to the window. Liza tried to laugh, too, as if she found the prank funny, but she couldn't even pretend. She hated wiggly, slimy things. Once again her younger siblings had proved how meek and pathetic she was. Charles and Emma were bigger, stronger, and braver. They knew they could tease Liza whenever they liked, without fear of retaliation.

Rolling over and pulling the quilt over her shoulders, Liza closed her eyes. She knew she should stand up to those two, but deep down, in the most private spot within her, she agreed with them. She was smaller than she should be, and she was filled with an infinite variety of fears. She just didn't quite measure up.



The next day, Liza woke with the light and made her yawning way downstairs, where Mama was stirring corn mush. Liza ladled the steaming mush into five bowls and set them on the table.

"Thank you kindly, Liza girl," her mother said, giving her a hug. "I can't believe my daughter will soon be 12. She is getting to be quite the young lady, isn't she, Papa?"

Liza's father carried a pail of water into the kitchen, pausing to wipe his feet before he stepped onto the clean floor. "She surely is," he said, giving her a warm smile.

Liza smiled back and began to eat her breakfast. Her parents moved into the parlor and murmured quietly. Liza suspected they were talking about her, but any hopes she had of successfully eavesdropping vanished as Charles, sneaking up behind her, slammed two pot lids together.

Her frightened shout caused both Charles and Emma to laugh uproariously. Emma put her arm around Liza and said, "Sorry, big sister, but you're such an easy target."

"I can't even hate them," Liza thought miserably. "They're not really mean; they're just mischievous." But she wished they would stop the constant teasing.

Liza went outside and squinted at the sky, where the winter sun was shining weakly. Soon the five of them would troop up Main Street, past the grocery store to Papa's blacksmith shop, then past the post office and the shoemaker's store to the school where Mama was a teacher and Liza, Charles, and Emma were students. Liza sat on the stoop to wait for the others.

"Oh, there you are." Liza's mother sat beside her on the stoop and lifted her face to the sun for a moment. Then she turned to Liza. "Papa and I have been talking, Liza. And we think you're old enough now to help me do something extremely important."

Liza's heart leaped, but she said nothing. Her mother went on in a quiet, urgent voice. "I want you to help me with Free Frank's lessons after school today."

Liza knew, as everyone did, that Free Frank paid her mother for reading and writing lessons. But she also knew that her mother did much more than that when she went to his farm.





Frank McWorter, or Free Frank as he was known, was their town's founding father. New Philadelphia was the first town in the United States that was planned and registered by a person freed from slavery. With determination and fortitude, Free Frank ran a farm and saved his money to buy his children's freedom from slavery, one by one.

"You children should be thankful to be living here," Liza's mother told them frequently. "You don't often see black folks and white folks living happily together in one place."

People in the town also knew that Free Frank had built his cellar with thick granite walls. That cellar had a secret room to hide people fleeing from slavery. It was a station along the Underground Railroad, which helped enslaved people escape north to freedom. The thick stone walls prevented tracker dogs from detecting the hidden people.

Liza's heart beat fast. "I'd love to help, Mama," she said.

CHAPTER 2

A NEW FRIEND

Liza tried to pay attention in class, but she was distracted by her excited anticipation. She was positive that her task that afternoon would have something to do with Frank's secret room.

After school, as she and her mother walked north toward Frank McWorter's farm, she held her tongue. She knew Mama would tell her what she needed to know when she needed to know it. Still, she couldn't help clearing her throat and gazing beseechingly at her mother.

Finally, Mama chuckled. "Okay, child, stop looking at me like that and I'll tell you." She took Liza's hand in her own. "You're old enough now to help with the fugitives," she said in a low voice.

Liza's heart skipped a beat, and she nodded eagerly.

"We're taking shoes to them today. Can you imagine traveling in bare feet in this winter weather? It's good that we have two kindhearted shoemakers in New Philadelphia. Anyway, these folks are on their way to freedom, but some of them may have to stay here for a while."

Liza asked why, but Mama just pointed at the farmhouse's cellar door and said, "You'll see very soon."

Liza followed her mother into the secret room behind the cellar wall. The room was dim, with only one candle shedding a feeble light. At first she thought it was empty.

But as her eyes grew accustomed to the gloom, she could make out ten people sitting silent and motionless around the walls.

"Don't worry," Liza's mother said reassuringly, "It's just my daughter Liza and me." It seemed as if they all exhaled at once, in a big gusty sigh of relief.



Her voice barely above a whisper, a woman sitting in the far corner said, "It's good to see you again, Annie." Liza couldn't help noticing that the woman was expecting a child, very soon by the look of her.

Liza's mother led her to the woman and addressed the girl sitting beside her. "Mary, this is my daughter Liza. She's 11, like you; right, Liza?"

Overcome with shyness, Liza immediately looked down at her feet, but her mother nudged her with a poke of her sharp elbow. Liza knew from bitter experience that her mother meant "Where are your manners, child?"

Liza looked up then and smiled at the other girl, who appeared to be short and thin, just like Liza herself.

Mary's face lit up and she smiled back, eyes sparkling. "I'm glad to meet you, Liza," she whispered.

And even as Liza whispered back that she, too, was happy to make the acquaintance, she had a strange premonition that this was a moment she would remember as long as she lived.

As Liza and her mother walked home that afternoon, Liza bubbled over with happiness. She told her mother that she was sure that she and Mary would be good friends. Her mother smiled but then spoke seriously. "Remember, you cannot talk about Mary to anyone, not even people from town. Never forget that you hold her life in your hands."

Liza bristled with indignation. "Of course I know that!" she said testily. As if she would risk the lives of any of the people being helped to freedom! "But we have so much in common. She even likes to read! She learned secretly with the children of the plantation owner."

"Then we'll have to bring her some books next time," her mother said. Then she shot her daughter a searching look. "What is it, child? What's troubling you?"

Liza had abruptly stopped walking and turned to her mother. "Somehow I never pictured someone who was born into slavery as being so very much like me. Why, Mary and I could be sisters." Tears burned her eyes, and a lump formed hard in her throat. "Why was I born free," she said, "while Mary must run and hide?"

Her mother wrapped her arms around Liza and squeezed her tight. "Oh, honey," she said. "Sometimes life makes no sense whatsoever. We just keep on doing the best we can."





After a few minutes, they resumed walking. "At least you will get to spend some time with Mary," Liza's mother said. "Ruth can't travel any further until she has her baby. It's a wonder she made it this far."

Liza skipped a little skip, delighted that she would be spending more time with her new friend.

Every day, Liza and her mother visited Ruth and Mary, bringing food, clothes, and books. While the women whispered in one corner of the hidden room, the girls rapidly became acquainted in another corner. Liza was amazed to learn how courageous her friend had been through all the rigors of her hard life and her dangerous flight to freedom.

One afternoon, Mary told Liza that they would be best friends forever, and Liza shook her head in wonder. "Why would you choose me as a friend? I'm small and weak and not at all disposed to bravery like you. I'm scared of just about everything!"

"Oh, Liza!" said Mary. "Of course I want you as my best friend. I already feel as if we're birds of a feather. Besides, there are all kinds of ways to be brave."

CHAPTER 3

EMERGENCY!

The next day the skies opened, unleashing torrents of rain and blasting wind. All day long, Liza feared they would not go out to the farm. As soon as school was let out, though, her mother said, "We have to go. Free Frank is away, helping a group to the next stop, so Ruth and Mary are alone out there."

It was a long, wet walk. When they finally reached the farm, Mary greeted them anxiously, wiping tears from her cheeks. "Mama's started having the baby," she said.

Liza's mother hurried over to Mary's mother. After a while, she took Liza aside and said, "We can't deliver this baby ourselves. Ruth needs the doctor, now!"

"I'll go, Mama!" Liza said.

Her mother gaped at her. "You can't," she said. "It's dark and stormy, and it's much too far back to town!"

Liza pulled her coat tight. "I have to do it, Mama."

Before her mother could say anything more, Liza kissed her on the cheek and then courageously stepped out into the wet, dark night.



The rain pounded, and the wind shrieked. It was the dead of night and so dark Liza could barely see her feet. She stumbled and fell into a mud puddle, but she was already so wet it didn't make much difference.

For one long moment, Liza lay there in the mud, in the rain, in the dark and all alone. "This is impossible," she thought, "especially for a small, weak girl terrified of the dark and horrified at what she can't see in the black night."

But she knew that if she went back, Mary's mother would still need help and that baby would still desperately require the doctor. She had to keep going.

"The hardest journey always starts with the first step," Liza told herself.

She rose shakily to her feet and continued toward town. Gradually she sped up, thinking of the danger that Mary's mother and the baby were in. She began to run, splashing through the mud and the driving rain. She must fetch the doctor quickly—it was a matter of life and death.

It was the longest, hardest trip of Liza's life. She was frightened every second as she made her way through the blackness, frozen and soaked to the skin. It seemed she would never reach town, never make it to the doctor.

And yet, despite it all, a new resolution filled her. She thought of how Mary, undaunted by obstacles, met each day with courage and good cheer. Couldn't she, Liza, be like her friend? Or at least pretend to be brave?

When lightning flashed, she forgot that she was always terrified of lightning. Thunder boomed, but she resolutely shrugged it off. She didn't have the spare energy to be frightened anymore; she needed every bit of strength for this last push.

At long last, Liza found herself on the way back to Free Frank's farm on the back of the doctor's horse. She wiped at her dripping head, but she wasn't thinking of the wet. "Please let us not be too late," she thought.

She led Dr. Smith into the cellar room behind the rock wall, and Mary's mother cried out in fear when she saw him.

Liza's mother soothed her, stroking her hand. "It's all right, Ruth. He's white, but he's a good man—like many other white people. He's not here to capture you; he's here to help you."



A baby boy was born safely that night, and they called him Henry. Soon the little family left on their trek north to Canada and freedom. Liza missed her best friend and wondered about Mary's family constantly.

Then in the spring, a letter came.

Dearest Liza,

All is well, and we are safe and free! Not a day goes by that we don't remember our brave and strong Liza. Thank you for your courage that saved Mama and little Henry.

With love from your best friend forever,

Mary

Drama

Mixed Messages

by Sharon Holt
illustrated by James Watson



Mc Graw Hill Education

PAIRED READ What Is Opera?

STRATEGIES & SKILLS

Comprehension
Strategy: Summarize
Skill: Theme

Content Standards
Social Studies
Culture and Diversity

Vocabulary
benefactor, empathy, endeavor, entail, extensive, indecision, multitude, tentatively

Word Count: 2,400**

Genre Drama

Essential Question

When are decisions hard to make?

Mixed Messages

by Sharon Holt
illustrated by James Watson

Act 1

Sibling Rivalry 2

Act 2

An Uncomfortable Feeling 8

Act 3

A Hard Decision 12

Respond to Reading 16

PAIRED READ **What Is Opera?** 17

Focus on Genre 20

ACT 1

SIBLING RIVALRY

CHARACTERS

STEVEN and SHEREE (twin brother and sister)
MOM
MR. STRANG (Sheree's teacher)
JASMINE (Sheree's friend)
OTHER STUDENTS
PRINCIPAL GRAY

SCENE 1

Scene: *The kitchen of a suburban home*
STEVEN *and* **SHEREE** *are eating breakfast.*

STEVEN: I hope you've sold a lot of cookies. Today's the last day to turn in the fundraising money.

SHEREE: (*pushes in her chair and picks up her breakfast dishes*) You know I'm not good at selling stuff—unlike you. You could sell grass to a farmer!

STEVEN: Very funny, but I'm not that good at selling. I just put in the extra effort because I want my class to win the camping trip for selling the most cookies. Don't you like camping?

SHEREE: I love camping. The thing is I could work day and night selling cookies, but it wouldn't make the slightest difference if the rest of my class didn't sell lots, too.

STEVEN: Wrong! If you worked that hard, you'd win the individual prize—and I know you like amusement parks.



SHEREE: I guess I just don't like them enough to work that hard for free tickets.

MOM *enters the room carrying the kids' backpacks. She sets them on the breakfast table.*

MOM: I thought you two would be ready by now. The bus will be here in ten minutes. Here's the money you've raised. Don't forget to hand it in.

MOM *takes two paper bags off the kitchen counter and puts one in each backpack.*



STEVEN: (*looks at the clock and stands up*) Actually, Mom, we were just talking about that. Sheree was telling me about her brilliant sales tactics. She's bound to win the competition.

SHEREE: (*picks up her backpack*) Oh, ha, ha.

MOM: Just go brush your teeth, please. I don't have time for sibling rivalry this morning.

SCENE 2

Scene: *Sheree's classroom*

MR. STRANG *stands at the front of the class.* **SHEREE**, **JASMINE**, *and the* **OTHER STUDENTS** *sit at their desks.*

MR. STRANG: May I have your attention, please, students? I have an announcement from Principal Gray.

JASMINE: (*leaning over and whispering to* **SHEREE**) I hope Mr. Strang is about to say there's no history test today. I totally forgot to study last night.

MR. STRANG: (*to* **JASMINE** *and* **SHEREE**) Pay attention, please, girls. You might be interested in what I'm about to announce. (*He picks up a piece of paper*) Here's Principal Gray's message. "Dear Room 8 students, you will be happy to hear that your class has raised an extensive amount of money toward the school's new gymnasium equipment. In fact, the amount raised has been sufficient to earn you first place …"

The class erupts into cheers.

MR. STRANG: (*raising his voice to be heard*) "… which is a class camping trip. Congratulations, Room 8."

JASMINE: Please, Mr. Strang, can we go to Yellowstone?

MR. STRANG: I haven't finished, Jasmine. (*turning back to the note*) "What's more, the person with the highest number of individual sales is also from Room 8." Drum roll, please!

SHEREE, **JASMINE**, *and the* **OTHER STUDENTS** *drum on their desks.*

MR. STRANG: The winner of three tickets to Hill Valley Amusement Park is our very own Sheree Bethell!



JASMINE *tries to high-five* **SHEREE**, *who is in shock. The class cheers.*

JASMINE: Yes! I knew it! We won! We won! Roller coasters, here we come!

MR. STRANG: (*raising his hands to get attention*) That's enough. Quiet down, please.

JASMINE *and the* **OTHER STUDENTS** *settle down.*

MR. STRANG: I think we have something of a celebrity among us. Stand up and tell us how you did it, Sheree.

SHEREE *looks around the room and then looks at* **JASMINE**, *who nods and smiles with encouragement.*

JASMINE: Go on, say something, or at least look happy!

SHEREE: (*standing up tentatively*) Ah, well, I … it wasn't really … I mean, I didn't even … I didn't sell …

MR. STRANG: Come on, Sheree. It's perfectly all right to celebrate when you've set a goal and achieved it.

SHEREE: You see, the thing is … I don't really know how … It's all a bit of a shock, but thanks anyway.

SHEREE *sits down.*

JASMINE: It's all right, Mr. Strang. She just can't believe it. Her twin brother is usually the one who wins all the prizes and gives the speeches.

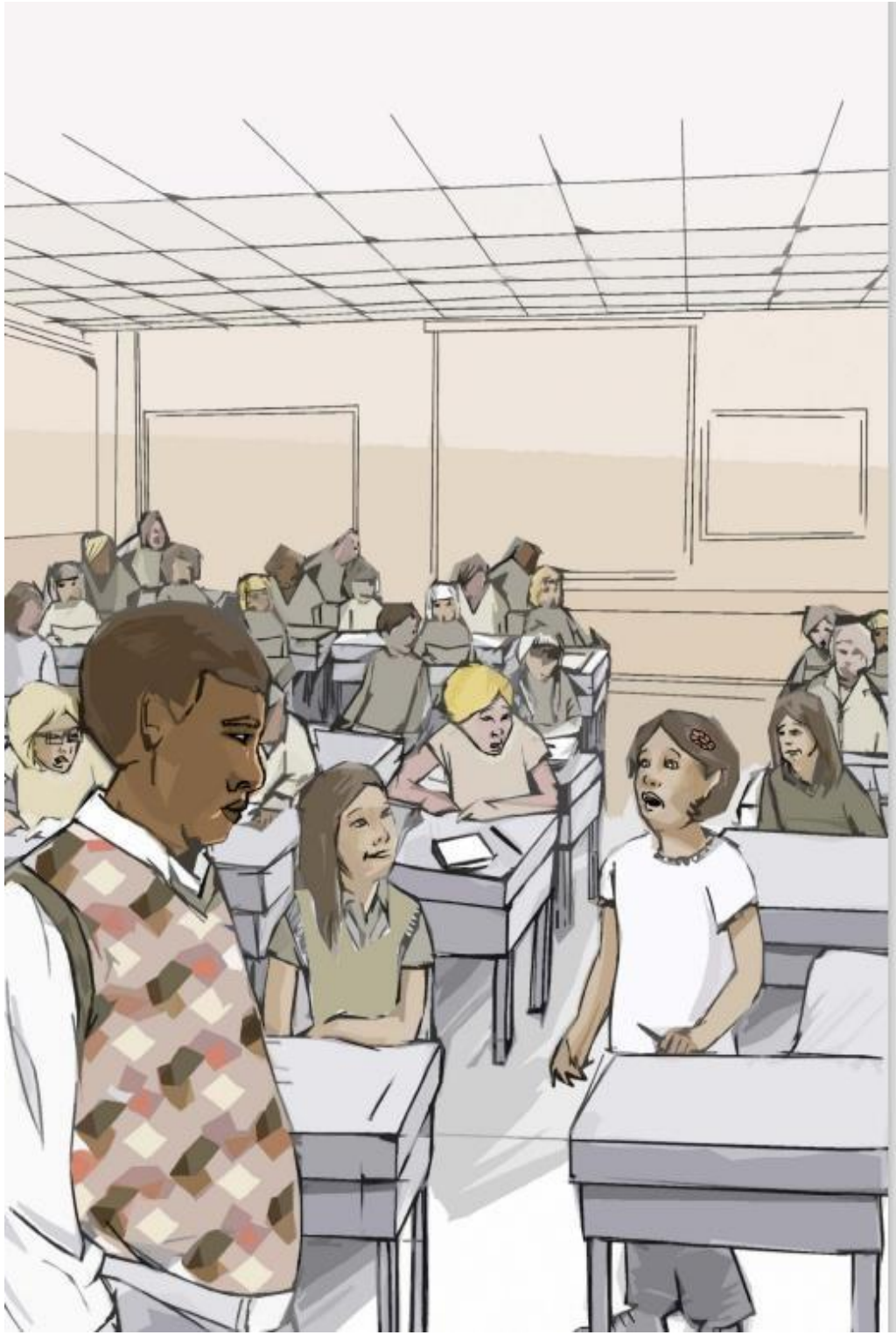
MR. STRANG: I understand. Congratulations on doing such a great job, Sheree. I'm sure there'll be plenty of suggestions for our camping trip, and my next endeavor is to find a solution that makes everyone happy. But right now, it's time for …

The bell rings to announce recess.

MR. STRANG: … recess.

STOP AND CHECK

Why is Sheree shocked to be named as the person with the highest sales?



ACT 2

AN UNCOMFORTABLE FEELING

SCENE 1

Scene: *The school hallway*

OTHER STUDENTS *are at their lockers.* **SHEREE** *and* **JASMINE** *are talking.*

JASMINE: That was such a surprise! I had no idea you sold so many cookies. Why didn't you tell me? Hey, did you see Jacob's face when Mr. Strang said you were the winner? He thought Billy and Divesh were his biggest competitors, and then you popped up from nowhere. Perfectly done! Aren't you excited?

SHEREE: It's so confusing. I just don't know what to do.

JASMINE: What's confusing? You won! We'll have an awesome time at the amusement park. Three tickets! You, me, and I guess your mom or dad, right? That is … if you want me to come.

SHEREE: Of course I do. It's just … (*sighing heavily*)

JASMINE: Everything's great. You worry too much.

SHEREE: You don't understand. It's all … I mean … this is such a mess. There's been a huge mix-up.

SHEREE *turns and grabs* **JASMINE** *by the shoulders.*

SHEREE: I can't possibly be the winner. I sold hardly any cookies at all! Almost none!

JASMINE: (*shocked*) What do you mean? How could that be? Is that why you were acting so strangely in class?

SHEREE: I guess so. I was just so confused about everything, but I know what to do now. I'll confess everything to Mr. Strang after recess. I'm sure he'll be able to find out who the real winner is.

JASMINE: You can't tell him. You can't tell anyone. It's not just about you anymore. It's about the whole class. Everyone's so excited about the camping trip.

SHEREE: Exactly. That's why I'm so confused. I want to do the right thing, I really do, but …

JASMINE: Maybe the right thing is keeping quiet. You can't let everyone down. If you say something now, the class will be devastated. Just forget about it, and no one will ever know.

SHEREE: I'll know! And it's not right. It's as if I've stolen the prize from someone who deserves it. How would you feel if you had sold all those cookies and someone else ended up walking away with the prize?

JASMINE: (*hands on hips*) You're making this way too complicated. We'll talk about it later. Now, let's get out of here before someone gets suspicious.

JASMINE *walks offstage,* **SHEREE** *follows reluctantly.*

SCENE 2

Scene: *Later in the school hallway*
STEVEN *is at the lockers, looking unhappy.* **SHEREE** *and* **JASMINE** *enter.*

JASMINE: What's up, Steven? Someone steal your lunch?

STEVEN: (*shaking his head*) No, much worse than that. Our class just found out that we didn't win anything in the cookie-selling competition. I wanted to go on that camping trip so much, and I worked so hard. Now I find out that it was all for nothing.

SHEREE: (*looking sad*) Oh, Steven. I'm really sorry.

STEVEN: It's not your fault. That'll teach me to spend all month dreaming about camping out under the stars.



JASMINE: The good news is our class won the prize!

STEVEN: (*surprised*) Really? That's … great news.

JASMINE: The even better news is that your sister …

SHEREE: (*points down the hallway*) Look, Steven. Mrs. Francis is standing over there with steam coming out of her ears. You don't want to get detention.

STEVEN *goes offstage, heading toward his classroom.* **MR. STRANG** *comes over to the lockers.*

MR. STRANG: I would like to get some teaching done today, girls. That will entail your actually being present in class.

MR. STRANG *and* **JASMINE** *go offstage.*

SHEREE: (*calling after them*) I'll be right there.

SHEREE *reaches into her backpack and sees the paper bag that had held the money for the cookies. She pulls it out. Her brother's name is on the bag.*

SHEREE: (*to herself*) Oh, no, Mom got the bags mixed up. I handed in Steven's money, and he handed in mine!



STOP AND CHECK

Why doesn't Sheree want Steven to know that she won the prize?

ACT 3

A HARD DECISION

SCENE 1

Scene: *The school auditorium*
SHEREE, **JASMINE**, **STEVEN**, *the* **OTHER STUDENTS**, *and* **MR. STRANG** *are attending assembly.* **PRINCIPAL GRAY** *is on the stage.*

PRINCIPAL GRAY: I am so grateful for the time and effort put in by everyone who worked so hard selling cookies. We can now purchase the new equipment for our athletic program. I am also grateful to our generous benefactor, the owner of High Street Sports, who has donated $5,000 of additional equipment.

Everyone claps.

PRINCIPAL GRAY: I can't tell you how proud I am to stand here in front of this multitude of hardworking students and give out some well-deserved prizes. Now, the student who sold the most cookies in the whole school is Sheree Bethell from Room 8. Well done, Sheree. Please come onto the stage.

JASMINE, *the* **OTHER STUDENTS**, *and* **MR. STRANG** *clap.* **STEVEN** *looks shocked.* **SHEREE** *walks onto the stage.*

SHEREE: (*takes a deep breath and speaks tentatively*) May I say something, please, Principal Gray?

PRINCIPAL GRAY: (*hands the microphone to* **SHEREE**) Of course, Sheree. The stage is yours.

SHEREE: (*takes another deep breath*) I was really surprised when I found out I won the competition. Unfortunately, I can't accept the prize. There was a mistake—my bag of cookie money was mixed up with my brother's bag. So Steven is the real winner, and he deserves it, too. He worked really hard. I apologize to everyone for this—especially for how my indecision over what to do has affected my class. I'm very sorry.

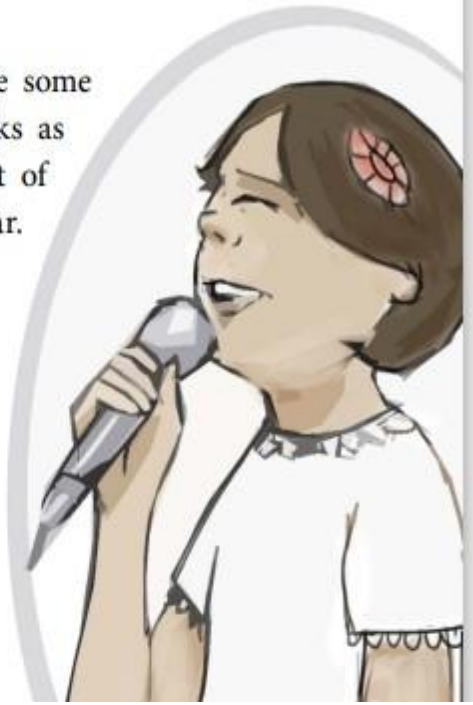
SHEREE *hands the microphone to* **PRINCIPAL GRAY** *and returns to her seat.*

PRINCIPAL GRAY: Well, thank you for your honesty, Sheree. It must have been very difficult to speak out and I'm sure we all have empathy for your situation. I am very impressed with how you have handled it.

The **OTHER STUDENTS**, **STEVEN**, **JASMINE**, *and* **MR. STRANG** *clap.*

PRINCIPAL GRAY: I have made some quick calculations, and it looks as if Mr. Strang's class had a lot of successful salespeople this year. So I'm happy to announce that Room 8 is still the winner of the camping trip.

JASMINE *high-fives* **SHEREE**, *and the* **OTHER STUDENTS** *cheer.*



SCENE 2

Scene: *A suburban street*
SHEREE *and* **JASMINE** *are walking home from school.*

JASMINE: I was looking forward to going to that amusement park. I would have screamed my lungs out on the roller coaster.

SHEREE: I know, but I wouldn't have been able to enjoy myself knowing that I hadn't been honest.

JASMINE: I don't think I would have been as honest as you were.

SHEREE: I think you would have. Imagine if your little sister had worked hard for weeks to sell all those cookies and somehow you got all the credit for it. Wouldn't you feel bad?

JASMINE: I guess so. I wouldn't want to do that to her.

STEVEN: (*calling from behind as he runs to catch up*) Hey, wait up. I have something to say to you.

SHEREE: I'm sorry, Steven. I should have spoken up sooner.

STEVEN: (*tries to look angry, then gives up and smiles*) Don't worry about it. I understand. By the time you realized what had happened, it was really hard to know what to do. I'm proud of you for getting up in front of the whole school and explaining things. That took guts.



SHEREE: Hey, thanks! And guess what? I asked Mr. Strang if you could join us on the camping trip, and he said yes.

STEVEN: Awesome! My friend Divesh is in your class. We'll have a great time. I can't wait.

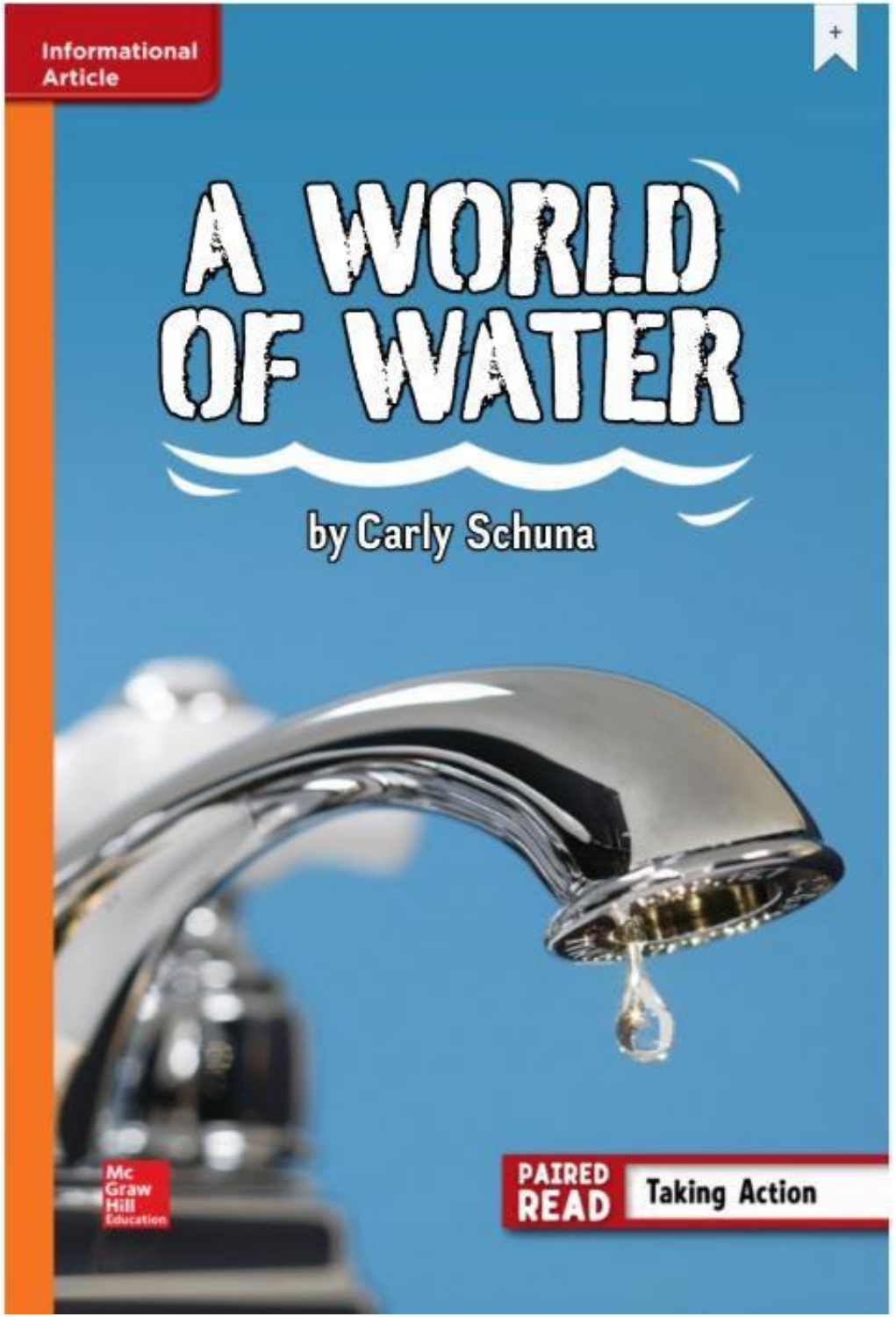
JASMINE: Oh, we love to create happy endings, don't we?

STEVEN: Actually, this one gets even better. How about you two take the amusement park tickets I won? It's not really my thing. I'd rather go camping any day.

JASMINE: Now, THAT'S what I call a happy ending!

STOP AND CHECK

How do Steven and Sheree end up getting what they wanted?



STRATEGIES & SKILLS

Comprehension

Strategy: Ask and Answer Questions

Skill: Main Idea and Key Details

Content Standards

Science

Earth Science

Vocabulary

advocates, commonplace, designate, initial, insulation, invasive, irrational, optimal

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**The total word count is based on words in the running text and headings only. Numerals and words in captions, labels, diagrams, charts, and sidebars are not included.

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Genre Informational Article

Essential Question

What steps can people take to promote a healthier environment?

A WORLD OF WATER

by Carly Schuna

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INTRODUCTION

Earth is both beautiful and fragile. All of our planet's resources need protection. The air we breathe and the water we drink are at risk.

The number of people on Earth is constantly growing. This has caused problems for the environment. One of the biggest problems is the shortage of clean water.

In many places, people can turn on the faucet to get clean, safe water. In these areas, water is commonplace. People might not think about where water comes from. However, there is a lack of clean water in many places around the world. Trying to get safe water to these places is very difficult.

Most drinking water comes from lakes, rivers, and groundwater. Groundwater is water from rain and snow that has soaked into the ground. Some places do not have enough drinking water. If there is water, it might not be safe to drink.

People and governments can help with this problem. People can help by conserving, or saving, water. Governments can pass laws to make sure people have **access** to clean water.



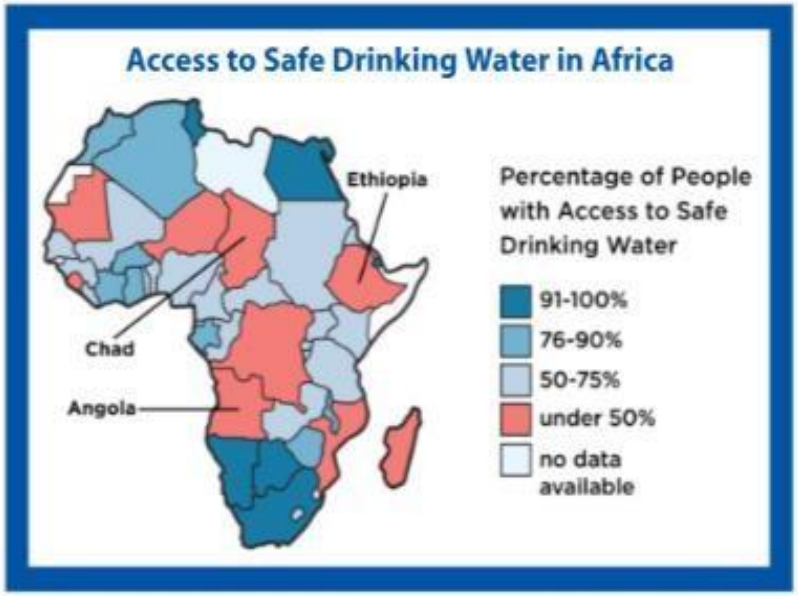
Water in rivers and lakes might not be safe to drink.

(bkgd) Ingram Publishing; (b) Photodisc/Getty Images

CHAPTER 1
DIRTY WATER

There are about 7 billion people on Earth. Almost 1 billion people do not have access to safe water.

Problems with safe water affect countries all over the world. The worst situations are in African nations such as Ethiopia, Chad, and Angola. Less than 50 percent of the people in these countries have clean water. Many people have no running water or plumbing in their homes. The only water they have is **contaminated**. They may have to walk long distances to get this water.



Water is unsafe in these countries for many reasons. Sometimes it contains animal or human waste. It might have other invasive elements, such as viruses or bacteria.

Nearly 500 children die in Africa every day from drinking unsafe water. Adults get sick from unsafe water, too. Sick adults can't take care of their families.

Many adults spend a lot of time finding and carrying water. Sometimes children have to help with this job, so they might not be able to go to school.



(Magni Ingram Publishing, By Lissa Harrison)

The lack of clean water means these women have to wash clothes in a muddy stream.

Digging a well is a simple way to give people clean water. A well draws water from deep in the ground. Wells provide insulation against the **pollutants** on Earth's surface.

A THIRST TO HELP

Even children can act as advocates for people who need clean water. In 1998, a six-year-old boy named Ryan Hreljac learned in school about the water problem in Africa. He wanted to help. Ryan's initial project was to raise the money needed to dig a well in Uganda. The well would provide a safe **source** of drinking water for the local people.

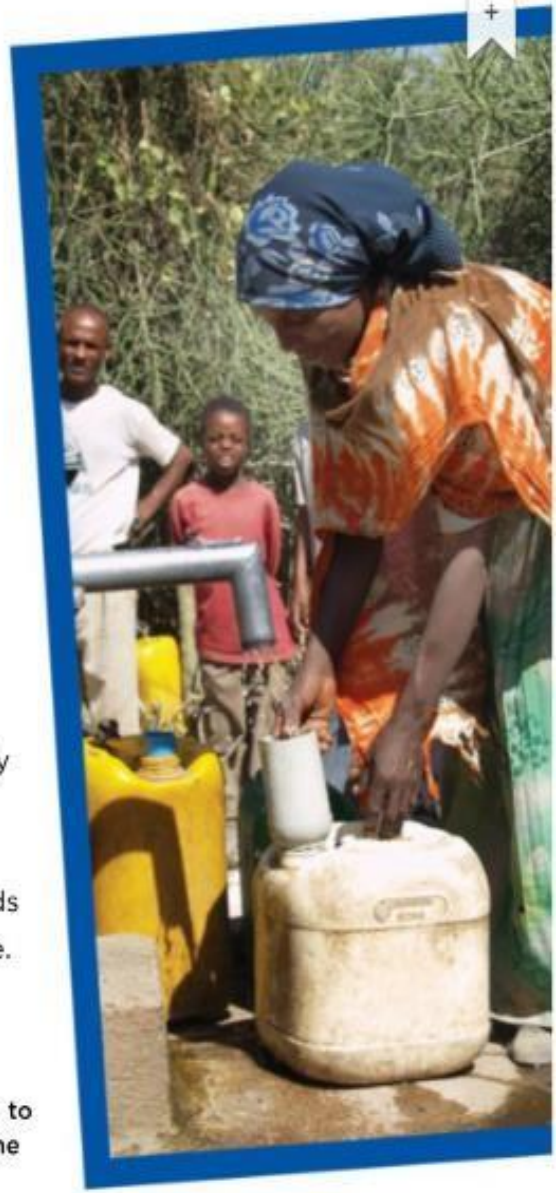


Ryan still works with his foundation to help people who don't have water.

Ryan's Well Foundation

Ryan did extra chores to raise money for the well. He asked others to designate money too. One year later, the well was built. The well is still providing clean, safe water to people in Uganda.

Ryan wanted to do more to help. He started a charity called the Ryan's Well Foundation. This charity has raised millions of dollars and brought clean water to hundreds of thousands of people.



Many people have access to clean water because of the Ryan's Well Foundation.

©Age Fotostock

STOP AND CHECK

Why is the water supply in parts of Africa unsafe?

CHAPTER 2

IN SHORT SUPPLY

Some people do not have access to clean water. Other people do not have any water at all. Water shortages are common in northern Mexico and the southwest of the United States. The climate there is hot and sunny. There are few large bodies of water and very little rain. Droughts, or long periods without rain, are common.

When an area does not receive enough water, governments import it from other places. This is not an optimal solution because importing water is difficult and expensive.

Drought is usually linked to other water **management** problems including climate change and pollution.

THE ROLE OF CLIMATE CHANGE

Climate change is a worldwide shift in weather patterns. There are changes to rainfall, temperature, and wind. Many scientists believe climate change is caused by gases in the atmosphere, such as carbon dioxide and methane. These are called greenhouse gases. Driving cars, cutting down forests, and other human activities produce greenhouse gases.

These gases build up in Earth's atmosphere. The gases trap the sun's heat. Over time this causes changes in the weather, including droughts.



POLLUTION PLAYS A PART

Climate change and drought have made water less plentiful in dry areas. They have made drinking water even scarcer. Some dry areas, such as northern Mexico and the southwest of the United States, have another problem. Their drinking water comes from polluted sources.

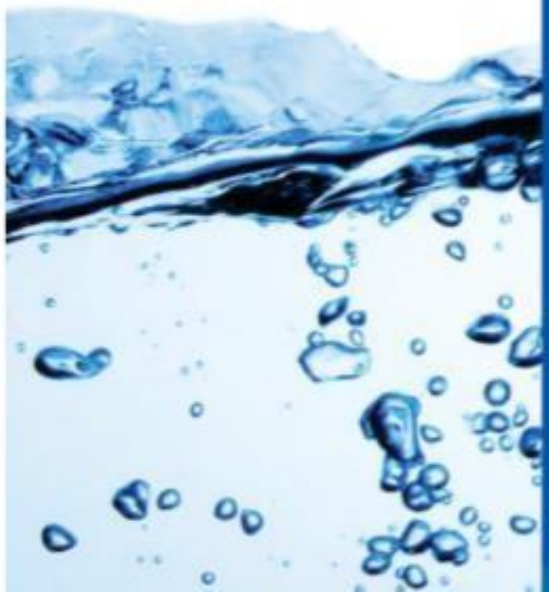
The Colorado River provides water for Las Vegas, Nevada, and for Phoenix, Arizona. Unfortunately, the river is polluted with human waste, toxic chemicals, and **radioactive** materials. The water is treated, but its quality is still poor. Drinking the water can make people sick.

How can people fix the problems caused by climate change, drought, and pollution? Desalination is one possible solution. This technique takes salt and other minerals out of sea water. This makes sea water safe to drink. Desalination is expensive. However, it might become cheaper in the future.

Governments can help, too. Laws can **regulate** how water is used. This can help conserve water.

STOP AND CHECK

How can climate change and pollution affect the water supply?





TAP WATER CONTAMINANTS

These contaminants, which are found in the Colorado River and other waterways, can cause many health problems.

ARSENIC: cancer; skin and circulatory problems

CADMIUM: kidney damage

CHROMIUM: skin problems and allergies

COPPER: stomach, liver, and kidney problems

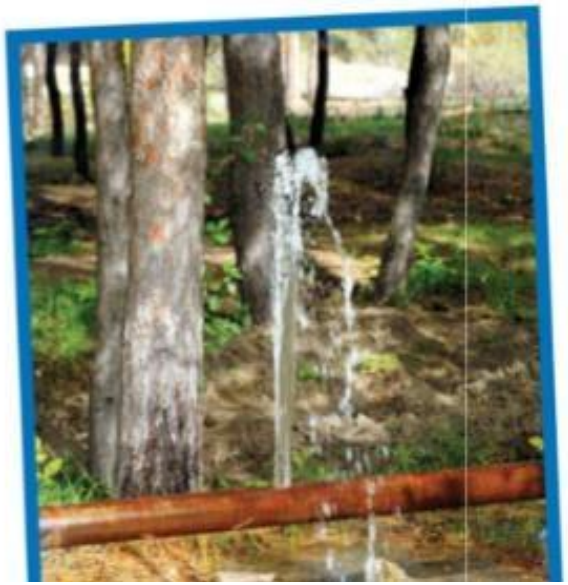
LEAD: kidney problems; high blood pressure

CHAPTER 3

WATER, WATER, EVERYWHERE

Water covers about three-quarters of Earth's surface. It seems that there should be plenty of water for everyone. However, most of this water is sea water. Only a small percentage is fresh water. Of this fresh water, only a small amount is safe to drink.

If we could **distribute** Earth's clean water equally around the world, it would save thousands of lives every year. People would not get sick from drinking dirty water. Unfortunately, some places have much more water than they need. Other places have much less.



Leaky pipes waste water.

There are ways to improve the supply of water. Repairing or replacing water **networks** that leak can save thousands of gallons each year.

Most people cannot replace or repair water networks on their own. However, they can talk to government officials. They can vote for improvements to how water is managed.

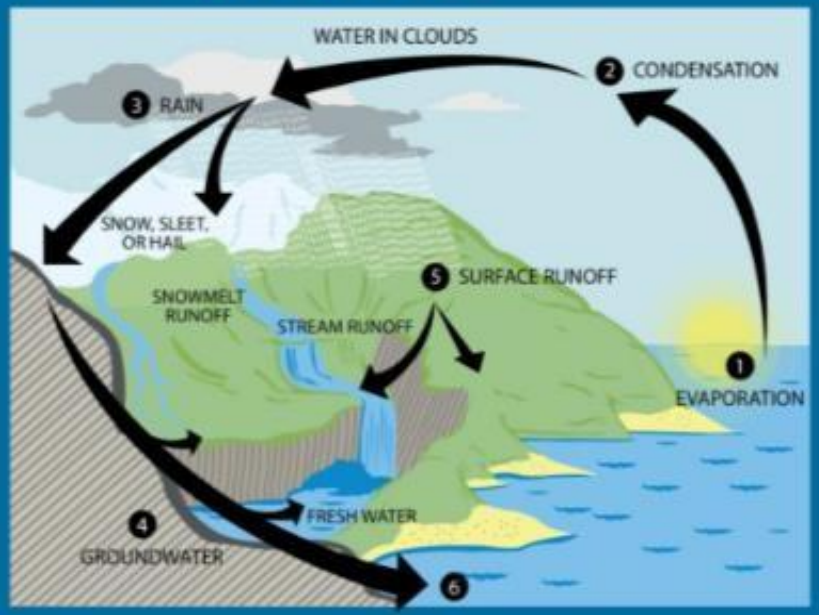
People can also use less water. Water flows from a tap at the rate of about 2.2 gallons per minute. People can save a lot of water by turning off the tap when they are brushing their teeth or soaping their hands.



Water treatment plants filter water and use chemicals to remove contaminants.

Cleaning up lakes and rivers can also help solve water problems. Water is constantly flowing. This means that contaminated water travels through the entire water cycle. It shows up again and again.

THE WATER CYCLE



1. Heat from the sun makes ocean water evaporate into the air.
2. Water in the air condenses into clouds.
3. The water falls to Earth as rain, snow, sleet, or hail.
4. On land some water seeps into the ground and is stored as groundwater.
5. Some water flows across the surface as runoff and is collected in streams, lakes, and rivers.
6. Water slowly finds its way back to the ocean again.



The pollution in this water will travel many miles downstream.

Pollution travels a long way. For this reason, people need to try to keep the water in their area clean.

There are easy ways that people can help with this. They can avoid dumping harmful products such as paint into sinks and drains. They can use environmentally safe household products. They can limit the use of fertilizers and pesticides. They can make sure not to litter in rivers, lakes, and oceans. By following these steps, people can help keep water clean and safe for everyone.

STOP AND CHECK

What can people do to help conserve the water supply in their area?

CONCLUSION

The problem of how to manage Earth's supply of clean water is serious, but it can be solved. The key is finding solutions that are affordable and long lasting.

Many methods can improve the water situation. Filtering, purifying, and importing water all help. Building wells helps, too. But Earth's population is growing, so the demand for water is increasing.

In the future, desalination may supply abundant safe water. There may be new ways to filter polluted water.

Many other methods for improving the water supply are being tested. One invention is called the Dew Bank. The Dew Bank can collect and store water from the atmosphere overnight. The device is modeled on the fog-basking beetle. This insect uses the ridges and furrows on its back to gather moisture from fog.

Water management is a complex environmental challenge. By following simple steps, we can all help work toward a solution.

THE WATER CRISIS AT A GLANCE

- One in eight people does not have access to safe drinking water.
- About 13 gallons (50 liters) of water a day are needed to meet an average person's basic needs.
- Some people in developing countries have access to only about 1 gallon (5 liters) of water a day, and it is often contaminated.
- Diseases related to contaminated water are one of the largest causes of death for children under age five in developing countries.



Dew collects on the fog-basking beetle's back and forms droplets the beetle can drink.